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Design and Fabrication of a Reliable Dual Recovery System for the N4 Rocket with Extended Telemetry Range

(FYP22-4)

Interim Report

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May 8, 2026

Declaration

We declare that this interim report is our original work and has not been presented for examination in any other institution.

We confirm that all sources of information, data, and published material used in preparing this report have been appropriately acknowledged and referenced in accordance with university academic standards.

We further affirm that this report reflects the progress made in our Final Year Project at this stage and that the work presented is a true account of the activities completed to date.

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Glenn Gatiba and Ian Joseph

May 8, 2026

Abstract

The Nakuja N4 rocket recovery system addresses three critical failure modes identified in previous missions: insufficient telemetry range, uncontrolled nichrome-wire burnout during parachute deployment, and absence of pre-flight electrical status feedback. This report outlines the design and development efforts undertaken to address these challenges.

The telemetry subsystem employs the XBee-PRO 900HP operating at 900 MHz in transparent Universal Asynchronous Receiver-Transmitter (UART) mode. A first-order link budget analysis confirms a +32 dB margin at 5 km using omnidirectional dipole antennas. The ignition subsystem introduces Pulse-Width Modulation (PWM)-controlled IRLZ44N Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) switching at 1 kHz, replacing the previous direct DC dump. Bench tests of 26 AWG nichrome elements demonstrated sustained energisation for approximately 30 seconds at minimum duty cycle, compared to burnout within 2 seconds under DC-equivalent drive, validating the controlled-heating approach. Additionally, a battery monitoring and continuity-detection circuit, centred on the ADS1115 16-bit Analog-to-Digital Converter (ADC), provides pre-launch voltage and igniter-state visibility. Mechanically, a stepped polyethylene terephthalate glycol (PETG) battery-carrier architecture accommodates four 18650 cells within the 95 mm avionics bay while routing dual threaded retention rods without interference.

Preliminary results indicate that all subsystem designs have been completed and verified through 2D/3D computer-aided design (CAD), EasyEDA printed circuit board (PCB) schematics, and Proteus circuit simulation. A custom flight-computer PCB is currently in fabrication, and 3D-printed structural parts are in production. Firmware modules for telemetry, sensor interfacing, PWM control, and flight-state management have been implemented on FreeRTOS.

The impact of these developments is significant: the Nakuja N4 system demon-

strates a robust and scalable approach to addressing critical recovery challenges, paving the way for enhanced reliability in future missions. The project will wholesomely improve how tests are done and

Contents

Nomenclature

Abbreviations

ADC Analog-to-Digital Converter

AGL Above Ground Level

BMS Battery Management System

CAD Computer-Aided Design

ESP32 Espressif 32-bit Microcontroller Platform

FEA Finite Element Analysis

GPS Global Positioning System

IMU Inertial Measurement Unit

LiPo Lithium Polymer Battery

MOSFET Metal-Oxide-Semiconductor Field-Effect Transistor

PETG Polyethylene Terephthalate Glycol

PWM Pulse Width Modulation

RF Radio Frequency

UART Universal Asynchronous Receiver-Transmitter

Symbols and Units

D Duty cycle, percent

f Frequency, hertz

I Current, ampere

P Power, watt

R Resistance, ohm

T Temperature, degree Celsius

V Voltage, volt

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1 Introduction

1.1 Background of Study

Rocket recovery systems represent the most critical subsystem for preserving flight data and hardware in high-power rocketry missions. The evolution of recovery systems has progressed from simple single-deployment methods to sophisticated dual-deployment architectures. Single deployment systems release a parachute at apogee, presenting risks of high drift radius and unpredictable landing zones. In contrast, dual-deployment systems utilise a drogue parachute at apogee for stability, followed by a main parachute deployment at lower altitude for controlled soft landing, establishing the standard for modern high-power rocketry operations [1, 2].

The Nakuja Project is a student-led rocketry programme based at Jomo Kenyatta University of Agriculture and Technology (JKUAT) and the Pan African University Institute for Basic Sciences Technology and Innovation (PAUSTI), operating in collaboration with the Kenya Space Agency (KSA) and JICA with the long-term goal of developing a liquid-propellant rocket capable of orbital payload delivery [3]. The programme has progressed through four rocket generations: N-1 (2021, simulated apogee 285 m), N-2 (2022, 500 m), N-3 (2024, 1,000 m target), and N-3.5 (May 2024, highest recorded apogee in the series at the Isiolo School of Infantry Range) [3]. Each generation has introduced lessons that inform the next design cycle.

The N-3.5 mission employed a piston-based single-deployment system using crimson powder and nichrome wire ignition, with telemetry transmitted over WiFi/MQTT via a Ubiquiti NanoStation [?]. While the launch was successful, post-flight analysis documented thermal damage to the charge holder, nichrome wire burnout after repeated pop tests, and effective telemetry range limited to the WiFi footprint of the NanoStation—constraints that were insufficient for the higher-altitude, longer-range N-4 mission profile [?, 10]. The N-4 rocket, targeting an apogee of 3,540 m as an entry into the Spaceport America Cup (SRAD/Solid/10,000 ft class), requires a recovery system capable of operating at significantly greater ranges and with improved deployment reliability [3].

1.2 Problem Statement

The N4 mission recovery system faces three documented critical challenges arising from N-3.5 field experience and the N-4 mission requirements:

1. **Telemetry range limitation:** The N-3.5 WiFi/MQTT-based system (Ubiquiti NanoStation) was constrained to line-of-sight WiFi range, which is insufficient for N-4's simulated apogee of 3,540 m and competition recovery distance requirements exceeding 5 km. Signal degradation during critical descent phases risks total telemetry loss and inability to locate the vehicle [?].

2. **Deployment unreliability:** Repeated N-3.5 pop tests documented nichrome wire burnout, thermal damage to the charge holder exceeding acceptable material limits, and inconsistent ejection from uncontrolled ignition energy. A single-deployment piston architecture also provides no redundancy against deployment failure at apogee [10].

3. **Absence of pre-flight system monitoring:** The N-3.5 system lacked real-time battery voltage and igniter continuity telemetry, preventing ground operators from verifying system readiness before launch and introducing risk of a failed deployment from an undetected low-battery or open-circuit condition [?].

These three gaps collectively threaten mission success through potential vehicle loss, deployment failure, and unverifiable launch readiness.

See Appendix F (Figure 32) for the N4 XBee telemetry architecture proposed to address gap 1.

1.3 Objectives

Main Objective: To design and fabricate a reliable dual recovery system for the N4 rocket with extended telemetry range.

Specific Objectives:

1. To extend the telemetry range of the recovery system beyond 5 km using an XBee-PRO 900HP radio link.
2. To improve deployment reliability through controlled PWM pyrotechnic ignition, targeting repeatable nichrome-wire energisation without wire burnout.
3. To design a rugged 3D-printed PETG avionics holder with vibration isolation, battery retention, and real-time ADS1115-based system diagnostics.
4. To validate the complete recovery system through ground pop tests and field range tests.

1.4 Justification

The development of a reliable recovery system is paramount to mission success. Loss of the rocket vehicle results not only in hardware loss but also complete loss of flight data critical for mission analysis and future design improvements. The N-4 rocket represents the Nakuja Project's most ambitious mission to date—a competition-class vehicle targeting 3,540 m—and the gap between its performance requirements and what the N-3.5 recovery system could support is substantial. The proposed improvements address documented failure modes from N-3.5 field tests and implement dual-deployment and long-range radio practices that are standard in the high-power rocketry community.

Beyond the immediate mission, the Nakuja Project's open-source, student-centred model means that validated recovery designs and documented lessons become resources for future engineering cohorts and for the broader African student rocketry community [3].

1.5 Expected Outcomes

The expected outcomes of this study are:

1. Telemetry range exceeding 5 km with reliable XBee-PRO 900HP link performance demonstrated during range tests.
2. Consistent and repeatable parachute deployment through PWM-controlled pyrotechnic ignition without nichrome wire burnout.
3. A fabricated 3D-printed PETG avionics holder providing vibration isolation, battery retention, and real-time ADS1115 system diagnostics.
4. A validated dual-deployment recovery system demonstrated through ground pop tests and integrated field testing.

2 Literature Review

2.1 Recovery Systems in Student Rocketry

Recovery systems preserve flight hardware and flight data and are therefore mission-critical in high-power student rocketry [1, 2]. A common practice is dual deployment: a drogue parachute at apogee for stability and reduced drift, then a main parachute at lower altitude for lower landing energy. Literature consistently shows that this staged strategy improves recovery reliability compared with single-deployment approaches in high-altitude missions [2].

The Nakuja Project's own mission progression illustrates this trend directly. N-1 (2021) and N-2 (2022) used basic single-deployment parachute ejection. N-3 (January 2024) introduced on-board flight logging and improved parachute reliability but retained a single-deployment architecture. N-3.5 (May 2024) advanced to a piston-based pyrotechnic ejection system but still deployed only a main parachute, with telemetry relying on a WiFi/MQTT link via a Ubiquiti NanoStation [?, 3]. The N-3.5 post-flight analysis documented thermal damage to the charge holder, nichrome wire burnout after repeated ground pop tests, and effective telemetry range constrained to WiFi coverage—all of which define requirements for the N-4 system [?, 10]. Figures 1 and 2 show the N-3 dual-deployment context.

Gap 1 — Dual-deployment adoption: Prior Nakuja missions did not implement staged dual deployment. N-3.5 documented that single-deployment piston ejection provides no redundancy at apogee, and that the absence of a drogue parachute increases drift and descent-rate uncertainty at high altitudes. No mission-specific data linking charge mass, ejection effectiveness, and dual-deployment sequencing for the N-4 airframe geometry exists in the Nakuja record.

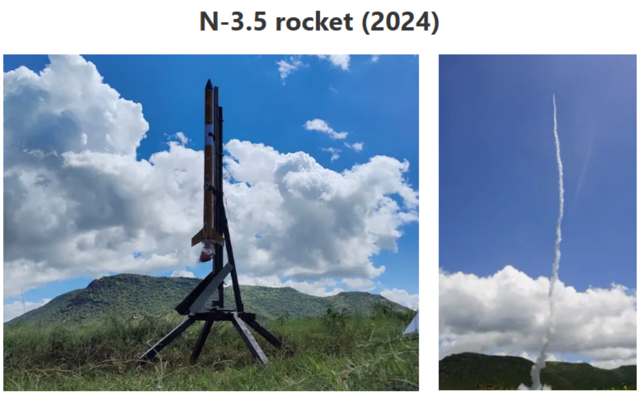


Figure 1: Nakuja N3 rocket with dual-deployment recovery configuration [3]

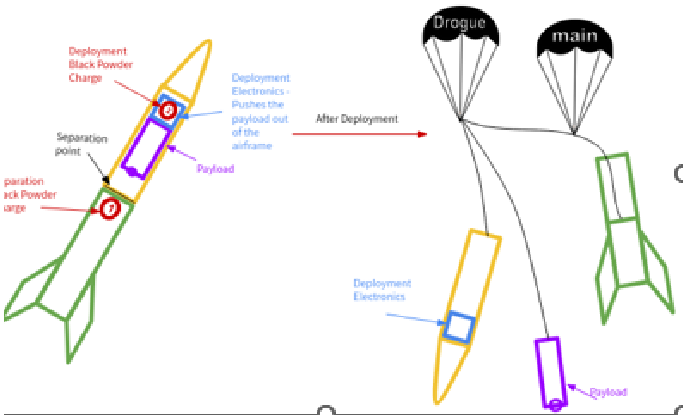


Figure 2: Nakuja dual-deployment flight during ascent [3]

2.2 Telemetry Systems Analysis

Table 1 compares telemetry options reported in student rocketry and related field systems.

Table 1: Telemetry System Comparison for Student Rocketry Applications

Technology	**Range**	**Data Cap**	**Strength**	**Limitation**
433 MHz RF	5–10 km	Very low	Simple, long range	No ACKs
Beacon	4 km/0.8 km	Low–Med	Simple	Range drops
LoRa SX1278	2–15 km	Low	Low power	Low BW
XBee 2.4 GHz	<2 km	Medium	Easy integration	Short range
XBee-PRO 900HP	**5–28 km**	**Med–High**	**Reliable, ACKs**	**Higher power**
ESP-NOW	<1 km	Medium	Low latency	Not long-range
WiFi/MQTT	<0.8 km	High	Easy integration	Network dep.
Cellular LTE	Global	High	Unlimited range	Network dep.

The N-3.5 system used WiFi/MQTT, which provides high data bandwidth but is limited to sub-kilometre range in open-field conditions without network infrastructure [?]. Range tests conducted during N-3.5 pre-flight preparations confirmed RSSI degradation beyond the NanoStation's effective beam, with intermittent packet loss at distances consistent with the N-4 expected recovery footprint [?]. Comparative studies show a tradeoff between range, bandwidth, protocol reliability, and implementation complexity [4]. 900 MHz-class links are favoured for long-range recovery because of better propagation under non-ideal field conditions compared with higher-frequency links [5]. Figures 3 and 4 summarise representative hardware and field RSSI behaviour.

Figure 3: XBee radio module used as the long-range telemetry transceiver platform [5]

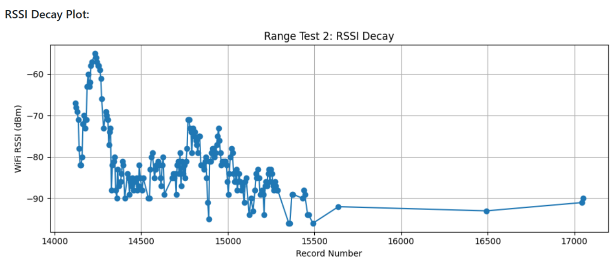


Figure 4: Beacon RSSI decay analysis showing signal degradation with distance and altitude. Project data.

Gap 2 — Telemetry range: The N-3.5 WiFi/MQTT architecture achieved sub-kilometre effective range, which is insufficient for the N-4 competition mission requiring reliable data recovery beyond 5 km. No Nakuja mission has yet validated a 900 MHz long-range link (XBee-PRO 900HP) for rocket telemetry.

2.2.1 Antenna Options in Recovery Telemetry

Literature and field practice identify several antenna families for rocket telemetry ground/air links: dipole and whip (omnidirectional, low complexity), patch and Bi-Quad (moderate gain, directional), and Yagi arrays (higher gain, tighter pointing requirement) [6, 7]. For dynamic flight profiles, antenna pattern behaviour and polarisation stability are as important as nominal gain. See Appendix F for representative antenna hardware (Figures 33 and 35) and dipole radiation pattern (Figure 34).

Published antenna fundamentals highlight two recurring operational issues in flight telemetry links: axis nulls in dipole-like radiation patterns and polarisation mismatch during vehicle attitude change [7]. These effects can dominate link performance even when radio modules have adequate power margins.

Gap 3 — Antenna selection guidance: Limited evidence-based antenna selection and orientation guidance exists for student recovery missions with dynamic vehicle attitude. No Nakuja documentation quantifies the performance difference between omnidirectional and directional ground antennas under real recovery descent scenarios, nor addresses polarisation mismatch during spin or tumble.

2.3 Pyrotechnic Deployment and Ignition Control

Black-powder and crimson-powder deployment remain common separation/ejection methods in student rocketry, but reliability depends on predictable charge energy, containment, venting, and material thermal tolerance [1, 8]. Literature distinguishes between commercial e-match igniters and custom nichrome approaches, with tradeoffs in repeatability, cost, and field test convenience [9]. Figures 5–8 show recurring deployment and igniter issues documented in Nakuja N-3.5 pop tests.

The N-3.5 recovery report confirms that nichrome wire was consumed in each

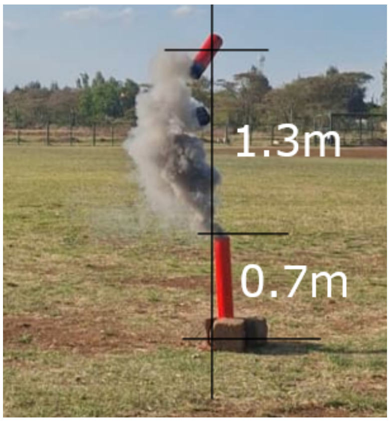


Figure 5: Excessive crimson powder deployment during N-3.5 field pop test [10]

Figure 6: Thermal damage on charge holder after N-3.5 pop test [10]

Figure 7: Post-deployment charring from containment leakage during N-3.5 ground test [10]

Figure 8: Cut nichrome wire after repeated pop tests, confirming single-use behaviour [10]

firing, requiring replacement before every test iteration [?]. For resistive igniters, literature reports that uncontrolled energy delivery causes premature wire failure or inconsistent ignition, while controlled-energy strategies—including PWM duty-cycle regulation—improve repeatability [9, 11, 12].

Combustion literature shows that short-duration pyrotechnic events are modelled as near-adiabatic transients for first-order chamber estimates, making thermodynamic pressure-volume curves essential for validating containment assumptions and charge-holder thermal boundaries [14, 15]. Figure 9 illustrates this relationship.

Gap 4 — Pyrotechnic predictability: Limited mission-specific data links crimson powder charge mass, chamber pressure behaviour, and repeatable ejection outcomes for the N-4 airframe geometry. No standardised chamber-seal verification procedure exists in the Nakuja record before live deployment tests.

Gap 5 — Ignition repeatability: Limited comparative data exists on e-match versus nichrome-based systems under repeated pre-flight ground-test cycles. N-3.5 confirmed nichrome single-use behaviour but did not quantify ignition-energy thresholds or controlled-duty-cycle strategies.

Gap 6 — Containment validation: Limited use of thermochemical adiabatic-

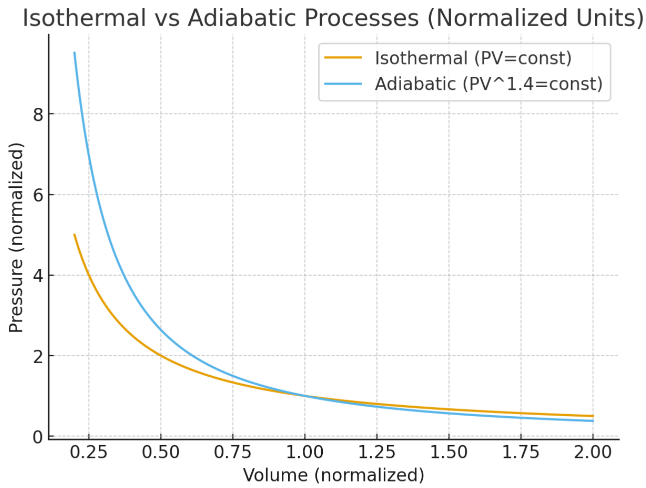


Figure 9: Isothermal and adiabatic pressure-volume comparison [14, 15]

curve evidence to justify contained explosion-cap deployment designs in Nakuja workflows. N-3.5 recorded a pressure transient consistent with 188 kPa during pop tests, but no charge holder was designed or validated against a derived burst/yield criterion [10].

2.4 Power and Structural Integrity in High-G Flight

Avionics for high-power rockets must survive sustained acceleration and impulsive shock during separation and deployment. Reported failure modes include connector intermittency, battery disconnect, solder fatigue, and sensor-noise coupling [16, 17]. Launch-load analysis for amateur rockets is framed from thrust-generated net force, where motor characteristics directly set structural loading during boost [18, 19].

The N-3.5 avionics bay transitioned from a wood-and-zip-tie N-3 design to a laser-cut perspex enclosure with 3D-printed battery supports [?]. This iteration improved compactness and component retention but was designed for the N-3.5 aluminium airframe with a 78.6 mm outer diameter; it does not fit the N-4 fiberglass airframe with a 109 mm outer diameter. Post-pop-test records documented pressure transients on the order of 188 kPa, reinforcing the need for mechanically robust holder designs and distributed fastening during deployment events [10]. Figure 10 shows the N-4 integrated avionics bay arrangement.

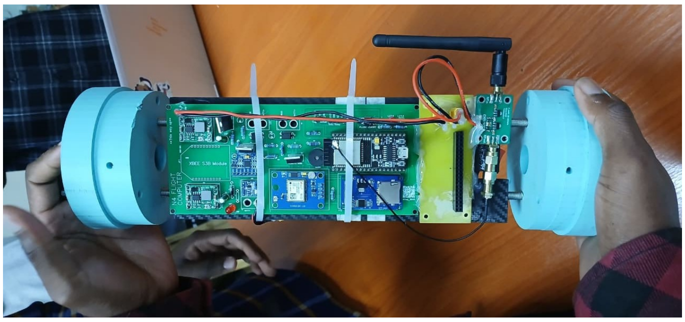


Figure 10: Assembled N-4 avionics bay with integrated battery retention and vibration isolation. Project fabrication.

2.4.1 PETG for Flight-Structure Components

Recent additive-manufacturing studies show PETG as a practical compromise for functional aerospace support parts because it offers improved ductility and impact tolerance relative to more brittle alternatives while retaining practical printability [20–23]. Structural integrity is strongly influenced by print orientation, infill strategy, and inter-layer bonding quality. This implies that material choice alone is insufficient; process quality and test validation are essential for reliable flight-support structures.

Battery-pack integration for common 18650-class cells contributes non-negligible mass at the module level, which increases inertial loading during launch and shock events and must therefore be included in holder-force calculations and mounting strategy design [24].

2.4.2 Flight-Computer Holder Design Progression

Literature and project practice emphasise iterative fixture refinement for avionics survivability in high-shock environments [16, 17]. The N-3.5 perspex bay offered improved retention over the N-3 wood design, but the shift to N-4's larger-diameter fiberglass airframe and 18650-based battery pack requires a full redesign. See Appendix F (Figure 36) for the earlier holder failure observed after pop tests, which confirms the need for shock-tolerant fixture geometry.

Gap 7 — Structural verification: PETG support structures and avionics sleds are commonly used in student rocketry but often lack launch and deployment shock validation for mission-specific loads. No validated PETG avionics holder exists for the N-4 109 mm airframe, 18650 battery configuration, or the 188 kPa pop-test pressure environment.

2.5 Pre-flight Monitoring and System Diagnostics

The N-3.5 flight computer included a buzzer that confirmed ESP32 WiFi connectivity and sensor detection at startup, but no real-time battery voltage or igniter-continuity telemetry was transmitted to the ground station [?]. Absence of such monitoring means operators cannot verify battery state-of-charge or continuity of the nichrome-wire circuit before committing to launch, introducing a direct path to deployment failure.

Industry guidance for high-power rocketry consistently recommends pre-launch continuity verification and launch-inhibit logic tied to power supply health as mandatory safety gates [16]. The N-3.5 documentation acknowledges this gap in its recommendations section and explicitly calls for addition of battery voltage monitoring in the next flight computer iteration [?].

Gap 8 — Pre-flight system monitoring: No Nakuja mission has implemented real-time ground-visible battery voltage or igniter-continuity telemetry with automated launch-inhibit logic. This gap is explicitly identified in the N-3.5

recovery report recommendations.

2.6 Summary of Research Gaps

Table 2 consolidates the eight gaps identified across the preceding subsections. Each gap is linked to the literature or project evidence from which it was derived and to the specific N-4 objective it motivates.

These eight gaps collectively frame the need for the methodology in Chapter 3, where design calculations, selection rationale, and implementation details are developed and validated against each gap.

Table 2: Summary of Research Gaps Identified from Literature and Nakuja Mission Evidence

#	Gap	Evidence Base	N-4 Objective Addressed
1	Dual-deployment adoption	N-3.5 used single-deployment piston; no dual-stage data for Nakuja airframes [?]	Specific Objective 1 (dual deployment)
2	Telemetry range	N-3.5 WiFi/MQTT limited to sub-km range; insufficient for 5 km+ competition recovery [?]	Specific Objective 1 (XBee 900HP link)
3	Antenna selection guidance	No evidence-based antenna selection for dynamic vehicle attitude in Nakuja missions [7]	Specific Objective 1 (antenna design)
4	Pyrotechnic predictability	No mission-specific charge-mass vs. ejection-effectiveness data for N-4 geometry [10]	Specific Objective 2 (controlled ignition)
5	Ignition repeatability	N-3.5 confirmed nichrome single-use burnout; no controlled-duty-cycle evaluation [?, 9]	Specific Objective 2 (PWM ignition)
6	Containment validation	No burst/yield analysis for charge holder; 188 kPa transient undocumented against material limit [10]	Specific Objective 2 (holder design)
7	Structural verification	No shock-validated PETG avionics holder for N-4 109 mm airframe and 18650 pack [10, 20]	Specific Objective 3 (avionics holder)

3 Methodology

This chapter presents the *what, why, and how* of the N4 recovery-system design process. In line with the interim-report guidelines, this chapter contains design considerations, calculations, circuit and CAD development workflow, and implementation planning for each subsystem.

3.1 Design Approach and Subsystem Decomposition

3.2 Methodology structure

The N4 recovery work follows an iterative design–build–test cycle. For clarity and traceability this chapter separates the lifecycle into two orthogonal views: (A) the project phases (linear progression through development and verification) and (B) the implementation modules (concurrent subsystem workstreams).

Project phases 1. Requirements definition — derive subsystem requirements from mission constraints and safety limits. 2. Concept generation — produce candidate mechanical, electrical and software solutions. 3. Concept selection — evaluate candidates against performance, manufacturability and safety. 4. Design and build — produce CAD, schematics and prototype hardware; capture PCB/EAGLE/EasyEDA sources. 5. Firmware implementation — develop modular ESP32 firmware and inter-module interfaces. 6. Validation and testing — staged bench and system-level tests, logging, and pre-flight checks.

Implementation modules These workstreams run in parallel across the phases above:

1. Telemetry and antenna strategy (XBee, beacon, ground-station),
2. Pyrotechnic deployment and ignition control (PWM, MOSFET drive),

3. Power and feedback instrumentation (ADS1115 monitoring, continuity sensing),

4. Mechanical containment and avionics support structure (holder, battery pack),

5. Embedded software and communication control (FSM, Communication-Manager).

This structure keeps the chapter descriptive ("what we did and why") rather than prescriptive, and maps directly to the deliverables referenced in Chapter 4 and the Appendices

3.3 Telemetry Module Design Methodology

3.3.1 Telemetry Module Scope

The telemetry module covers flight-to-ground data transmission hardware, antenna strategy, communication reliability constraints, and integration interfaces to the ESP32 flight computer firmware [5, 25, 26].

3.3.2 Telemetry Design Considerations

Design considerations for telemetry selection are:

- Minimum operational range of at least 5 km line-of-sight.
- Robust packet delivery with low drop rate in dynamic flight attitude.
- Simple and deterministic integration with ESP32 UART telemetry output.
- Field-deployable ground-station setup without complex infrastructure.
- Compatibility with future communication redundancy (beacon/MQTT fallback).

3.3.3 Telemetry Parts List

Table 3: Telemetry Module Parts List

Part	Specification	Function in Module
XBee-PRO 900HP (S3B)	900 MHz ISM, up to +24 dBm Tx power	Primary long-range telemetry radio for rocket and base station links
ESP32 Flight Computer	Dual-core MCU, UART interfaces	Generates and streams telemetry packet data to the radio interface
900 MHz Duck Dipole Antenna	Approx. +2 dBi omnidirectional gain	Baseline antenna on rocket and ground for initial long-range tests
XBee Shield/UART Interface	Transparent serial routing	Hardware adaptation between ESP32 UART and XBee module pins

3.3.4 Telemetry Design-Option Comparison

3.3.5 Chosen Telemetry Design and Justification

The selected telemetry hardware is the **Digi XBee-PRO 900HP** operating in transparent UART mode because it provides strong range performance, straightforward ESP32 integration, and robust field operation with minimal protocol complexity [5, 25, 26].

3.3.6 Communication Mode Architecture and the Beacon Protocol

The N4 flight computer implements three independent communication channels managed by a software CommunicationManager [25]: (1) MQTT over WiFi for close-range pad operations, (2) the 900 MHz XBee link for primary long-range recovery telemetry, and (3) a *beacon protocol* currently being researched as a redundant fallback channel.

Table 4: Telemetry Design Comparison Against Module Considerations

Option	**Range Capability**	**Integration Effort**	**Reliability Potential**	**Assessment vs Requirements**
XBee 2.4 GHz	Low–medium	Low	Medium	Insufficient long-range margin for target mission envelope
LoRa (generic modules)	Medium–high	Medium	Medium–high	Good range but added integration and protocol adaptation overhead
ESP-NOW/Beacon only	Medium (field dependent)	Medium	Medium	Useful backup path but not preferred as sole primary long-range channel
XBee-PRO 900HP	**High (5 km+ target)**	**Low–medium**	**High**	**Best match to range, robustness, and implementation constraints**

What is the Beacon protocol? A beacon is a raw 802.11 management frame injected directly into the IEEE 802.11 (2.4 GHz WiFi) radio medium without establishing a conventional network association. The rocket does not connect to any access point; instead, the ESP32 radio injects custom management frames — each carrying the telemetry CSV payload in a vendor-specific information element — directly into the air at a user-defined interval. A ground-station ESP32 running in promiscuous (*sniffer*) mode captures all frames that match the payload magic number regardless of association state, and extracts the RSSI directly from the hardware receive descriptor [27]. The return path (ground to rocket) uses ESP-NOW unicast packets, which provide confirmed delivery without requiring a broker or access point.

The beacon system has been field-tested to 4 km line-of-sight using a 2.4 GHz duck antenna and inline signal amplifier on the rocket side, and a directional patch antenna with amplifier at the ground station [27]. It is employed as the **redundant research channel** — the XBee 900 MHz link remains the primary mission telemetry path. The CommunicationManager supports automatic fallback from MQTT to beacon when WiFi connectivity is lost, and allows runtime switching between all three modes via serial, MQTT, or ESP-NOW commands [25]. The telemetry architecture is shown in Appendix **??** (Figure 32).

3.3.7 Antenna Options and Selection Using Link-Budget Math

Candidate antennas considered for the telemetry link are summarized in Table 5.

The selected baseline antenna pair for current implementation is the 900 MHz duck dipole. Its feasibility is checked with the first-order link budget. Free-space path loss (FSPL) models the signal attenuation over distance in the absence of obstacles [5, 6]:

$$\text{FSPL}\,(\text{dB}) = 20\log_{10}(d) + 20\log_{10}(f) - 147.55 \tag{1}$$

where d is the link distance in metres, f is the carrier frequency in hertz, and

Table 5: Antenna Options for Telemetry Module

Antenna Type	**Typical Gain**	**Pattern**	**Methodology Assessment**
Duck Dipole (900 MHz)	∼2 dBi	Omnidirectional	Baseline choice due to simplicity and tolerance to vehicle rotation
Patch Antenna (900 MHz)	∼7 dBi	Directional	Higher gain but stricter pointing and build tolerance requirements
Bi-Quad (900 MHz)	∼10–11 dBi	Directional	Considered as a base-station upgrade candidate to improve null-zone robustness

147.55 is the SI-unit normalisation constant derived from the $\lambda/(4\pi R)$ free-space propagation factor [6].

At the N4 target range of $d = 5{,}000$ m and centre frequency of $f = 915 \times 10^6$ Hz, the free-space path loss is 105.66 dB, which is within the expected range for 900 MHz ISM-band communications at this distance [5]. Received power at the ground station is obtained from the link-budget equation [5, 6]:

$$P_{rx} = P_{tx} + G_{tx} + G_{rx} - \text{FSPL} \tag{2}$$

where P_{tx} is the transmit power in dBm, G_{tx} and G_{rx} are the transmit and receive antenna gains in dBi, and FSPL is the free-space path loss calculated above (dB). The XBee-PRO 900HP rated output is $P_{tx} = +24$ dBm [5], and both ends use duck dipole antennas with $G_{tx} = G_{rx} = 2$ dBi. From equation 2 the received power at the ground station, P_{rx}, was obtained as -77.66 dBm.

Link margin quantifies the safety buffer above the minimum detectable signal threshold [6]:

$$\text{Margin} = P_{rx} - P_{sens} \tag{3}$$

where P_{sens} is the XBee-PRO 900HP rated receiver sensitivity (-110 dBm at 10% packet-error rate [5]). Using the values above gives a link margin of 32.34 dB, which leaves enough headroom for atmospheric loss, multipath fading, and small misalignments. The antenna pattern discussion is continued in Appendix **??** (Figure 34).

To mitigate dipole null effects at the ground segment, a **Bi-Quad antenna is being considered as a potential base-station alternative** for improved directional gain and better coverage across critical tracking sectors [6, 7].

3.4 Electrical Module with Deployment Submodule Methodology

3.4.1 Electrical + Deployment Module Scope

This module covers onboard signal interfacing, telemetry electrical routing, power/continuity monitoring, and pyrotechnic actuation electronics. It is paired with the deployment submodule because ignition-drive design depends directly on electrical interface limits and power budget constraints.

3.4.2 Design Considerations

Key electrical/deployment design considerations are:

- The flight computer has limited available pins, requiring protocol-level optimization between UART and I^2C paths.
- Telemetry radio (XBee) requires hardware serial, with remapped RX/TX pins on ESP32 to preserve other interfaces.

- Power monitoring and chute-state sensing require additional channels beyond native MCU pin allocation, motivating ADC expansion over I^2C.
- The electronics stack must fit inside the 95 mm avionics bay and align with holder mounting geometry and rod clearances.
- The deployment channel is validated against a 16.8 V fully charged 4S LiPo input and a 14 V minimum arm threshold so the sensing and inhibit logic cover both the worst-case supply peak and the lowest safe pre-fire condition.
- Pyro switching must be compatible with 3.3 V logic drive and sustain current pulses without excessive heating.

3.4.3 Protocol and Interface Option Comparison

Table 6 compares interface options against pin, telemetry, and monitoring requirements.

Table 6: Electrical Interface Options for Pin-Constrained Flight Computer Integration

Interface Option	**Pin Usage**	**Suitability for XBee**	**Suitability for Monitoring**	**Assessment**
UART-only for all peripherals	High	Excellent for XBee stream	Poor channel scalability	Not optimal for mixed telemetry + sensing architecture
I^2C-only for all peripherals	Low	Poor for high-rate transparent radio stream	Good for ADC expansion	Not suitable as sole interface due to XBee serial requirements
Hybrid: UART + I^2C (chosen)	**Balanced**	**Best for XBee hardware serial**	**Best for ADS1115 expansion**	**Best fit for pin limits and subsystem coexistence**

3.4.4 Chosen Electrical Architecture

The selected architecture uses hardware UART for XBee telemetry and I^2C for ADS1115-based monitoring expansion. This preserves deterministic serial throughput for radio transmission while offloading analog channel scaling to an external converter integrated on the existing I^2C bus [5, 25, 26, 30, 31].

3.4.5 Dimensional Constraints and Power Requirements

Table 7 summarizes packaging and electrical limits that bound the design.

Table 7: Electrical Module Constraints and Design Power Budget

Constraint/Paramete	**Design Value**	**Implication**
Avionics bay diameter envelope	95 mm	PCB and daughter-board placements constrained by cylindrical packaging
PCB footprint target	up to 70 mm × 70 mm	Allows routing margin while preserving service clearance
XBee supply rail	3.3 V (radio domain)	Requires clean low-voltage rail and UART-level compatibility
Main pyro bus	14–15 V domain	High-energy actuation path isolated from logic rail
Chute line deployment potential	16.8 V max input, 14 V minimum arm threshold	Monitoring divider and ADC scaling must tolerate the full operating window

3.4.6 Electrical Materials List

Table 8 lists the selected components and their roles in deployment control.

Table 8: Electrical and Deployment Submodule Materials

Component	**Key Rating/Interface**	**Role in Selected Design**
ESP32 flight computer	3.3 V logic, UART + I^2C	Core control and communication processing
XBee-PRO 900HP	UART transparent mode	Long-range telemetry radio interface
ADS1115 ADC	16-bit, I^2C interface	Voltage and chute channel expansion for pin-limited MCU
IRLZ44N MOSFET	Logic-level gate, low $R_{DS(on)}$	Pyro-channel low-side switching for PWM ignition
Voltage divider network	47 kΩ / 10 kΩ	Scales 14–15 V monitoring signals to safe ADC range
Nichrome igniter element	Resistive heating load	Converts electrical pulse to ignition heat
BMS module	Over-voltage/under-voltage protection	Caps and protects battery bus for controlled deployment operation

3.4.7 Circuit Development Workflow

Electrical implementation follows a staged concept-to-production flow:

1. Breadboard and wiring validation,
2. Fritzing-level integration drafting,
3. EasyEDA schematic and PCB capture,

4. Bench verification and revision control.

The Fritzing layouts below document the early wiring verification stage. Final EasyEDA schematics and fabricated PCB designs are presented in Chapter 4.

See Appendix **??** (Figure 38).

See Appendix **??** (Figure 39).

3.4.8 Power-Monitoring Calculations (ADS1115)

This subsection describes the ADS1115 battery-monitoring and chute-state sensing design. The divider is sized for a 4S LiPo upper limit of 16.8 V and a 14 V minimum arm threshold so the monitoring and inhibit logic cover the full safe operating window. The monitoring path uses a 4.7 kΩ/1.1 kΩ voltage divider into the ADS1115 [31]. A voltage divider scales an input voltage proportionally using two series resistors [32]:

$$V_{ADC} = V_{in} \cdot \frac{R_2}{R_1 + R_2} = V_{in} \cdot \frac{1.1}{4.7 + 1.1} \tag{4}$$

where V_{in} is the battery bus voltage to be measured, $R_1 = 4.7$ kΩ is the upper divider resistor, $R_2 = 1.1$ kΩ is the lower resistor, and V_{ADC} is the scaled output voltage presented to the ADC pin.

For $V_{in} = 16.8$ V:

$$V_{ADC,16.8} = 16.8 \cdot \frac{1.1}{4.7 + 1.1} \approx 3.186\,\mathrm{V} \tag{5}$$

For $V_{in} = 14$ V:

$$V_{ADC,14} = 14 \cdot \frac{1.1}{4.7 + 1.1} \approx 2.655\,\mathrm{V} \tag{6}$$

The ADS1115 is configured at a ±4.096 V full-scale range (FSR) with 16-bit resolution (32,768 counts over the positive half-range) [31]. The corresponding digital code is:

$$N = \frac{V_{ADC}}{4.096} \times 32{,}767 \tag{7}$$

where N is the 16-bit ADC output count and 32,767 is the maximum positive code $(2^{15}-1)$. For the two design points:

$$N_{16.8} \approx 25{,}487 \quad (\text{at } V_{in} = 16.8\,\text{V}), \qquad N_{14} \approx 21{,}242 \quad (\text{at } V_{in} = 14\,\text{V}) \tag{8}$$

These ranges provide clear monitoring margin below ADC full-scale while preserving sensitivity to deployment-state transitions.

3.4.9 Deployment PWM and MOSFET Selection

The pyro channels are driven through PWM-controlled MOSFET stages to avoid nichrome burnout and improve reusability across repeated ground tests [11, 12, 33, 34].

(a) Nichrome wire element characterisation:

The igniter elements use 26 AWG nichrome resistance wire with a linear resistance of approximately 8.5 Ω/m [9]. For a representative bridge-wire element of length $l_e = 0.10$ m (10 cm), the igniter element resistance is:

$$R_{elem} = \rho_{NiCr} \cdot l_e \tag{9}$$

where ρ_{NiCr} is the linear resistance per unit length (Ω/m) and l_e is the active element length (m). Substituting values:

$$R_{elem} = 8.5 \times 0.10 = 0.85\,\Omega \tag{10}$$

At a target drive voltage $V_{target} = 6.0$ V, steady-state current in the element is given by Ohm's Law [32]:

$$I_{elem} = \frac{V_{target}}{R_{elem}} \tag{11}$$

Substituting:

$$I_{elem} = \frac{6.0}{0.85} \approx 7.06\,\text{A} \tag{12}$$

Corresponding instantaneous resistive power dissipation:

$$P_{wire} = \frac{V_{target}^2}{R_{elem}} \tag{13}$$

Substituting:

$$P_{wire} = \frac{6.0^2}{0.85} \approx 42.4\,\mathrm{W} \tag{14}$$

Using PWM to limit the effective voltage below the ignition threshold until arm state is confirmed prevents unintended heating on the bench. Bench characterisation with 26 AWG elements showed that at minimum PWM duty cycle the wire withstood approximately 30 seconds of sustained energisation without failure, compared to burnout within 2 seconds at maximum (DC-equivalent) duty. Pyrotechnic ignition and pop tests have not yet been conducted and are scheduled for the next project phase.

(b) PWM duty-cycle model:

PWM controls the average voltage seen by the nichrome wire by rapidly switching the MOSFET on and off. The average output voltage is proportional to the duty cycle [12, 33]:

$$D = \frac{V_{target}}{V_{supply}} \tag{15}$$

where D is the dimensionless duty cycle (0 to 1), V_{target} is the desired average element voltage, and V_{supply} is the LiPo supply voltage. For the nominal case $V_{target} = 6.0$ V and $V_{supply} = 15.0$ V:

$$D = \frac{6.0}{15.0} = 0.40 \quad \text{(i.e. 40\% duty cycle)} \tag{16}$$

(c) IRLZ44N vs IRF-type MOSFET choice:

The selected switch is IRLZ44N (logic-level gate-drive compatible) rather than a standard IRF-series part, because the ESP32 gate output is 3.3 V and logic-level devices achieve low on-state resistance at reduced gate voltage [35, 36]. Conduction loss in a MOSFET switch is [36]:

$$P_{cond} = I^2 R_{DS(on)} \tag{17}$$

where I is the drain current and $R_{DS(on)}$ is the drain-to-source on-resistance. The IRLZ44N is rated $V_{DSS} = 55$ V, $I_D = 47$ A, with $R_{DS(on)} = 22$ mΩ at $V_{GS} = 5$ V [35]. For a representative $I = 3$ A pulse through a logic-level switch ($R_{DS(on)} \approx 0.022\,\Omega$):

$$P_{cond,IRLZ} = 3^2 \times 0.022 \approx 0.20\,\mathrm{W} \tag{18}$$

For a standard non-logic-level part at low gate drive ($R_{DS(on)} \gtrsim 0.05\,\Omega$):

$$P_{cond,IRF} \gtrsim 3^2 \times 0.05 \gtrsim 0.45\,\mathrm{W} \tag{19}$$

The lower dissipation and reliable 3.3 V gate compatibility motivate selecting IRLZ44N for the deployment channels.

3.4.10 Additional Implementation Details

To complete the electrical-deployment pairing, the design also includes:

- Hardware serial pin remapping for XBee RX/TX on ESP32 without blocking I^2C monitoring.
- Segregated low-voltage logic and high-energy pyro routing on PCB for reduced coupling.
- Pre-arm validation of battery/chute channels before enabling deployment PWM outputs.

3.5 Structural (Mechanical) Module Design Methodology

3.5.1 Structural Module Scope

The structural module defines the flight-computer holder, battery-carrier geometry, threaded-rod integration, mounting-hole allocation, and interface provision for the XBee extension board and battery management system (BMS). Beyond individual component retention, the holder functions as the primary mechanical bed that fastens the complete battery pack as a unified assembly and provides the upper-platform mounting surface for the XBee extension board and beacon antenna module.

3.5.2 Structural Design Considerations

The structural design constraints are:

- Fit within the 95 mm avionics-bay envelope.
- Survive launch and deployment shock/vibration events (design baseline: up to 4g sustained/inertial loading, with higher short-duration pop-test transients).
- Package four 18650 cells in parallel while avoiding rod interference.
- Maintain two threaded rods at 50 mm spacing and approximately 6 mm offset from the rear holder reference edge.
- Raise edge battery seats relative to mid seats so outer cells clear rod paths.
- Allocate and distribute self-tapping screw locations for load sharing.
- Provide mounting provision above battery holders for XBee extension and BMS.
- Include threaded holes for brass columns used to mount the flight computer.

3.5.3 Structural Parts List

Table 9: Structural Module Parts List

Part	Nominal Quantity	Function in Structural Module
Flight-computer holder body (PETG)	1	Primary load-carrying printed structure supporting electronics and battery fixtures
18650 cells	4	Parallel energy source mounted in dedicated mid/edge battery carriers
Mid battery holders (shorter)	2	Support central cells while preserving assembly height and alignment envelope
Edge battery holders (raised)	2	Raise outer cells to clear threaded-rod trajectories
Threaded rods	2	Structural retention members at 50 mm spacing and 6 mm rear offset reference
Self-tapping screws	multiple	Distributed fastening points for force sharing and anti-vibration retention
Brass columns (standoffs)	4	Flight-computer mounting through threaded holes in the holder
XBee extension mount + BMS mount area	1 each	Allocated upper-plane features for electronics integration above cell carriers

3.5.4 Structural Design Option Comparison

Table 10: Structural Design Comparison Against Mechanical Requirements

Option	**Envelope Fit (95 mm)**	**Shock/Vibrati Robustness**	**Assembly/Int**	**Assessment**
Single-level battery plane	Good	Medium–low	Simple	Rod interference risk for edge cells; weaker shock retention
Adhesive-only retention	Good	Low	Simple initially	Poor serviceability and limited repeatability under vibration/pop tests
Stepped holders + rods + screws	**Good**	**High**	**Moderate**	**Best balance of fit, force distribution, maintainability, and subsystem integration**

3.5.5 Chosen Structural Design and Implementation

The selected design is a stepped battery-holder architecture with raised edge supports, shorter mid supports, dual threaded-rod retention, and distributed self-tapping screw points. This combination provides geometric clearance and better load sharing under launch and pop-test disturbances. The holder additionally fastens the complete 4-cell battery pack as a single retained unit, and the upper platform above the battery carriers serves as the mounting bed for both the XBee extension board and the beacon antenna module, keeping all avionics electronics within the 95 mm bay envelope.

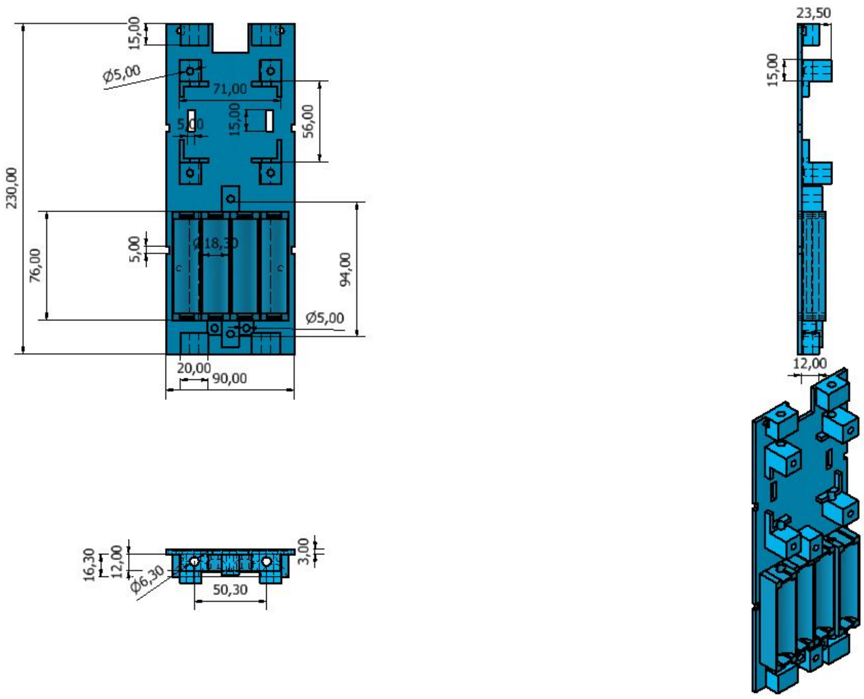


Figure 11: Flight-computer holder sled: stepped battery seating, load path, and upper electronics mounting platform

The sled fastens the complete 4-cell pack as a unified assembly and serves as the mounting bed for the XBee extension board and beacon antenna above the battery carriers.

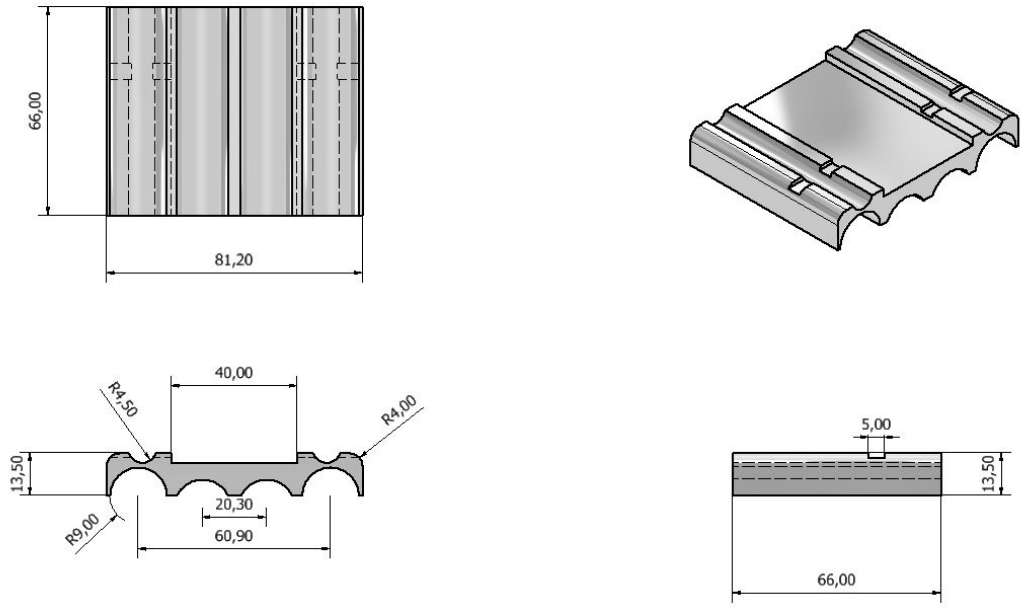


Figure 12: 18650 battery case for the stepped carrier assembly

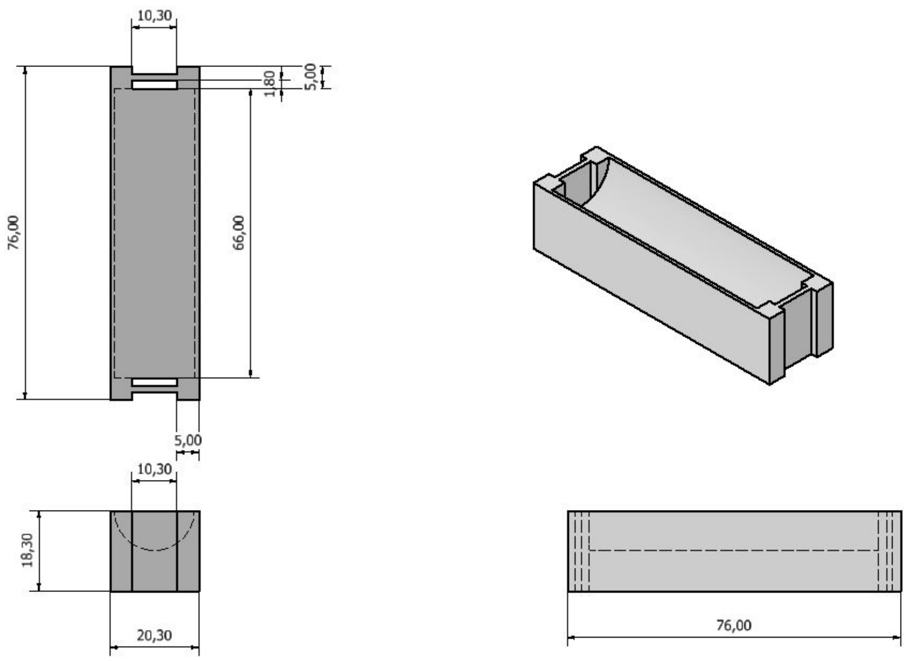


Figure 13: Raised edge battery support clearing the threaded-rod path

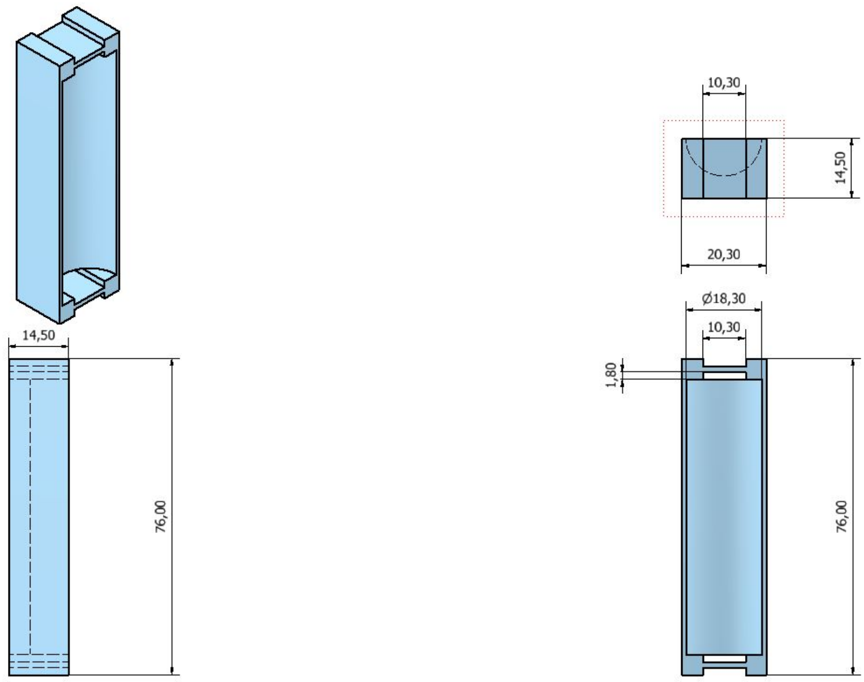


Figure 14: Mid battery support for controlled pack height and fit

3.5.6 Structural Load Calculations

(a) 18650 battery mass and weight Using a representative 18650 cell mass of $m_{cell} \approx 0.047$ kg (typical of high-capacity 18650 datasheets [24]), total pack mass is:

$$m_{pack} = n_{cells} \cdot m_{cell} \tag{20}$$

where $n_{cells} = 4$ is the number of cells. Substituting:

$$m_{pack} = 4 \times 0.047 = 0.188\,\text{kg} \tag{21}$$

Static gravitational load from the four cells:

$$W_{pack} = m_{pack} \cdot g \tag{22}$$

where $g = 9.81$ m/s^2 is the standard gravitational acceleration. Substituting:

$$W_{pack} = 0.188 \times 9.81 \approx 1.84\,\text{N} \tag{23}$$

(b) Launch inertial force (4g design case) The inertial load on the battery pack during launch is obtained from Newton's second law [18, 19]:

$$F_{inertial} = m_{pack} \cdot a_{launch} \tag{24}$$

where a_{launch} is the peak launch acceleration. For a 4g design envelope:

$$a_{launch} = 4g = 4 \times 9.81 = 39.24\,\mathrm{m/s^2} \tag{25}$$

Substituting:

$$F_{launch} = 0.188 \times 39.24 \approx 7.38\,\mathrm{N} \tag{26}$$

(c) Pop-test pressure-driven force (188 kPa reference) Pressure force on a surface is [10]:

$$F = P \cdot A \tag{27}$$

where P is the applied pressure (Pa) and A is the projected area (m^2). The previously recorded pop-test overpressure reference is P_{pop} = 188 kPa [10]. The conservative projected area uses the full 95 mm bay section:

$$A_{95} = \pi \left(\frac{0.095}{2}\right)^2 \approx 7.09 \times 10^{-3}\,\mathrm{m}^2 \tag{28}$$

Substituting:

$$F_{pop} = P_{pop} \cdot A_{95} = 188{,}000 \times 7.09 \times 10^{-3} \approx 1.33 \times 10^3\,\mathrm{N} \tag{29}$$

(d) Impulse estimate for pop transient Impulse is the product of force and the duration over which it acts:

$$J = F_{pop} \cdot \Delta t \tag{30}$$

where J is impulse (N·s) and Δt is the pressure-pulse duration. For a representative short pressure-pulse duration Δt = 10 ms:

$$J \approx 1{,}330 \times 0.01 \approx 13.3\,\mathrm{N \cdot s} \tag{31}$$

(e) Load distribution through fasteners and supports For distributed fastening with n screw/support points, first-order load per point under a total applied force F is:

$$F_{point} = \frac{F}{n} \tag{32}$$

where n is the number of fastener points. Under $F_{pop} \approx$ 1,330 N: 4 points carry $\sim$ 332 N each; 8 points carry $\sim$ 166 N each. This demonstrates why distributed fastening reduces local stress concentration and improves structural reliability.

3.6 Embedded Software Methodology

3.6.1 Software Module Scope

The software module governs all firmware executing on the ESP32 flight computer, covering three implementation areas: (i) telemetry and XBee communication management, (ii) pyrotechnic deployment state-machine control via PWM, and (iii) power and continuity monitoring integration. Area (iii) is in active development and has not yet been validated against live hardware at the time of this report.

3.6.2 Software Design Considerations

Three core problems drive the software design:

1. **XBee Integration:** The XBee-PRO 900HP must stream telemetry reliably at a deterministic data rate using hardware serial. Interface selection (SPI vs UART), data rate constraints, and pre-flight radio configuration via XCTU must all be resolved before firmware integration.

2. **Pyrotechnic Deployment via PWM:** The deployment state machine must safely arm, fire, and confirm chute deployment events using PWM-gated MOSFET drive without false actuation or premature triggering under vibration or electromagnetic interference.

3. **Power Monitoring Integration:** Real-time battery voltage and chute-channel continuity feedback from the ADS1115 must be incorporated into pre-flight arm logic and appended to the outbound telemetry packet. This integration is not yet complete in the current firmware build.

3.6.3 XBee Communication Interface: SPI vs UART

The XBee-PRO 900HP hardware exposes only a UART serial interface for transparent-mode data transfer; it does not offer SPI or I^2C bus modes [5]. SPI is therefore inapplicable, and the design uses the ESP32 hardware UART peripheral with remapped RX/TX GPIO pins, preserving all other peripheral assignments on the flight computer [30].

Key serial parameters governing the XBee-to-ESP32 link are listed in Table 11.

Table 11: XBee-PRO 900HP Serial Interface Parameters

Parameter	**Value**	**Implementation Note**
Default baud rate	9600 baud	Factory default; must be updated via XCTU `BD` command
N4 firmware baud rate	115200 baud	Reduces per-packet latency; set identically on flight and ground units
Over-air RF data rate	200 kbps	Fixed by module hardware; not configurable from firmware side
Interface type	UART, 3.3 V logic	No SPI path; transparent mode passes raw bytes directly to/from ESP32
Telemetry packet interval	100 ms (10 Hz)	Frame rate used in N4 build; well below over-air throughput ceiling

3.6.4 Radio Pre-Flight Configuration via XCTU

XCTU is Digi's official configuration and test utility for XBee modules. The N4 pre-flight procedure programs both flight and ground-station units via XCTU before each integration cycle [5]:

1. **Network ID (ID):** Set identically on both units (e.g., `7FFF`) to isolate the N4 network from other XBee traffic in the field.
2. **Baud Rate (BD):** Set to 115200 on both units to match firmware UART initialization.
3. **Channel Mask:** 900 MHz ISM band frequency-hopping profile configured to minimize ambient interference.
4. **Transmit Power (PL):** Maximum (+24 dBm) for full downlink link-margin.
5. **API Mode (AP):** Set to 0 (transparent mode) so the module passes raw bytes directly without packet-framing overhead, which suits the simple CSV/binary telemetry stream.
6. **Sleep Mode (SM):** Disabled; module remains active continuously during all flight phases.

After configuration, a loopback range test confirms bidirectional packet delivery before avionics bay assembly. Module profiles are saved in XCTU for rapid re-flashing if hardware replacement is required before launch.

3.6.5 Communication Manager Architecture

All three channels are coordinated by the CommunicationManager class, which selects and combines active transmission modes at runtime [25]. The mode-selection logic and fallback flow are shown in Appendix **??** (Figure 40).

See Appendix **??** (Figure 40).

3.6.6 Firmware State Machine Architecture

The firmware is structured as a finite-state machine (FSM) executed inside the ESP32 main `loop()`, with telemetry transmission running continuously across all states via hardware serial interrupt. The full state-machine diagram is presented in Appendix ?? (Figure 41).

3.6.7 Pyrotechnic Deployment Software Flow

The deployment channel is driven only when the FSM reaches the APOGEE or RECOVERY state and all pre-fire checks pass. See Appendix ?? (Figure 42) for the full gating logic moved to the appendix.

Key firmware constants for the deployment channel [33, 34]:

- `PWM_DUTY_IGNITE = 102` ($D = 0.40$, $V_{\text{eff}} \approx 6$ V from 15 V supply)
- `LAUNCH_INHIBIT_V = 14.0` V minimum battery voltage for arm enable
- `CHUTE_COUNT_THRESHOLD = 1000` ADS1115 ADC counts for continuity present
- `PWM_FREQ = 1000` Hz LEDC timer frequency for the ignition channel

3.6.8 Power Monitoring Software Integration (In Progress)

Battery voltage and chute-channel continuity are read from the ADS1115 over I^2C and back-calculated to real-world voltage using the inverse of the hardware voltage divider [31]:

$$V_{battery} = V_{ADC} \times \frac{R_1 + R_2}{R_2} = V_{ADC} \times \frac{57.0}{10.0} \tag{33}$$

where V_{ADC} is the raw ADC reading in volts, and the factor 57/10 is the reciprocal of the hardware divider ratio [32].

The monitoring flow is presented in Appendix ?? (Figure 43).

The reconstructed voltage is appended to the telemetry packet and compared against the inhibit threshold. If the value falls below `LAUNCH_INHIBIT_V`, the deployment arm flag is cleared and a "`PWR INHIBIT`" status byte is broadcast over XBee to the ground station before launch [33]. The hardware concept for this monitor is shown in Appendix **??** (Figure 44).

Current integration status: The hardware design and Proteus simulation of the ADS1115 monitoring circuit are complete (detailed in Section 3, Electrical Module). Firmware I^2C polling of ADS1115 channels and the voltage reconstruction function are coded but have not yet been validated against the assembled PCB. Integration testing is planned as the next milestone before launch readiness review.

3.7 Validation and Test Method

Testing is staged from component to full-system level:

- **Component level:** PWM-output verification, continuity checks, and ADC calibration.
- **Subsystem level:** telemetry range trials, ignition repeatability tests, and enclosure leak/charring checks.
- **System level:** integrated ground rehearsal with full communication stack and deployment command-path validation.

4 Results and Discussion

This chapter presents the results obtained from design, fabrication, and testing activities completed to date. Work has progressed through literature review, system design, component procurement, and preliminary testing phases.

4.1 System Design Completion

4.1.1 Mechanical Structural Design

The structural design follows the guideline requirement of presenting dimensioned 2D drawings first, followed by 3D CAD models and assembly views.

2D Dimensioned Drawings Figures **??**–**??** present the dimensioned 2D orthographic views for the battery carrier components. All units are in millimetres.

See Appendix **??** (Figure 45).

See Appendix **??** (Figure 46).

See Appendix **??** (Figure 47).

3D CAD Models and Assembly The 3D CAD models confirm dimensional fit, threaded-rod routing, and sub-assembly relationships within the 95 mm avionics bay.

See Appendix **??** (Figure 48).

See Appendix **??** (Figure 49).

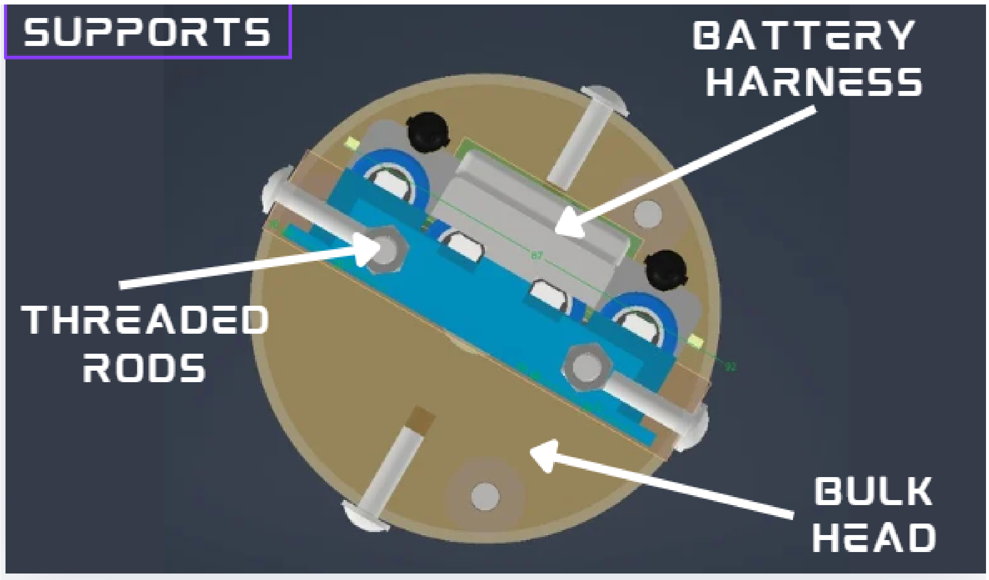
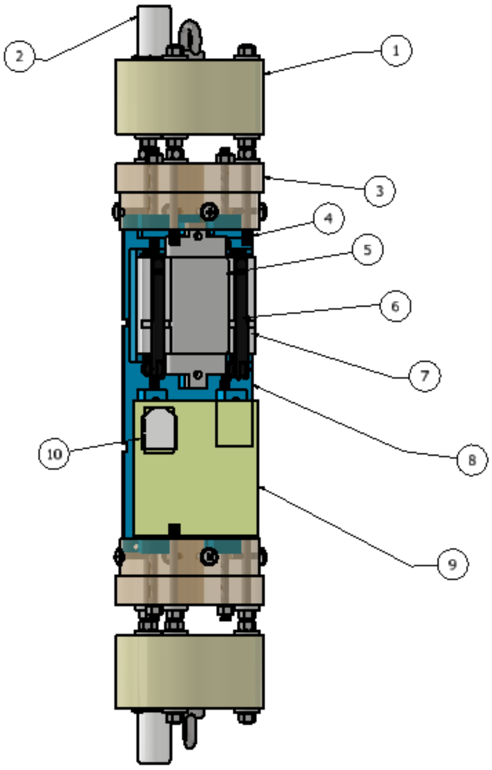


Figure 15: Labelled cross-sectional avionics bay showing component stacking and clearances



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	2	Coupler	
2	2	Charge Holders	
3	3	Bulk_Head	
4	2	Threaded Rods	
5	1	Battery Harness	STEP AP242 Edition 2
6	2	Antenna	
7	1	Battery cover	
8	1	FC Sled	
9	1	Xbee Extension Circuit	
10	1	xbee	

Figure 16: Full avionics bay assembly parts list with component identification

See Appendix **??** (Figure 50).

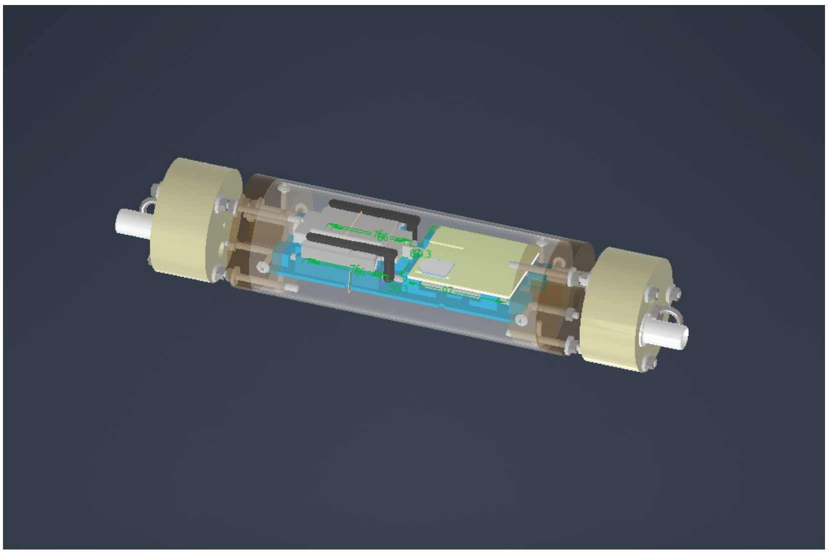


Figure 17: Full avionics bay integration in cylindrical airframe section

4.1.2 Electrical Circuit Design

EasyEDA schematics for the flight computer and base station were developed from the Fritzing layouts presented in Chapter 3. The flight computer schematic is included in Appendix B. Figure ?? shows the base station circuit schematic.

See Appendix ?? (Figure 51).

4.1.3 Power Monitoring Simulation

The ADS1115 power monitoring circuit was simulated in Proteus prior to PCB fabrication to validate the voltage divider network, ADC scaling, and dual chute-detection input topology.

See Appendix ?? (Figure 52).

Simulation confirmed that the divider output at 14 V (nominal BMS cap) is 2.456 V, and at 15 V (deployment line transient) is 2.632 V, both well within the ADS1115 $\pm$4.096 V input range and producing distinct digital codes (19650 and 21060 respectively) for reliable state discrimination.

4.1.4 Fabricated PCB

The custom flight-computer PCB was fabricated using a chemical etching process. The board accommodates the ESP32 DevKit, XBee UART interface, MOSFET ignition channels, and ADS1115 monitoring circuit. The fabricated PCB measures 88 mm $\times$ 93 mm, which fits within the 90 mm $\times$ 100 mm space allocated for the circuit in the avionics bay design.

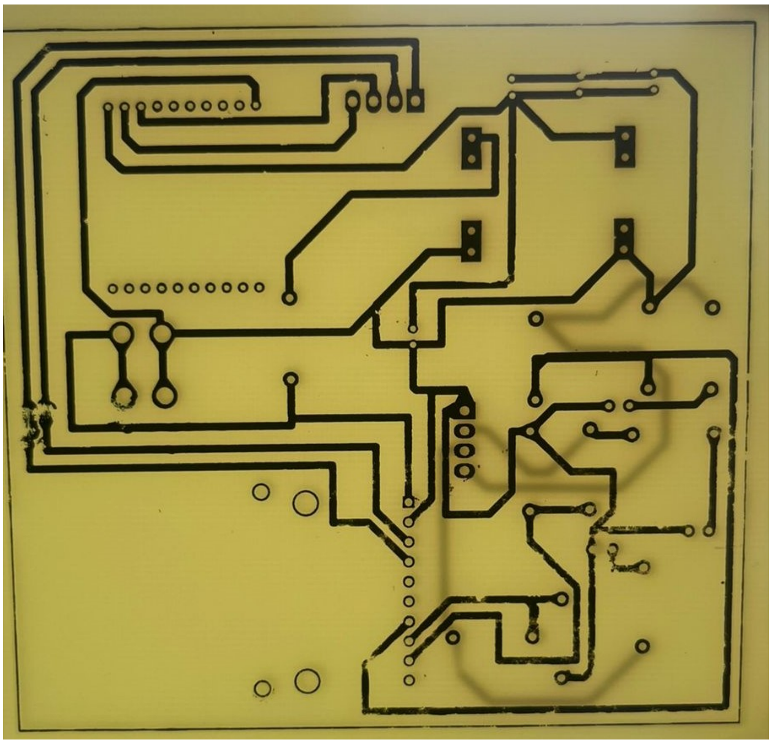


Figure 18: Flight-computer PCB after chemical etching, bare board stage

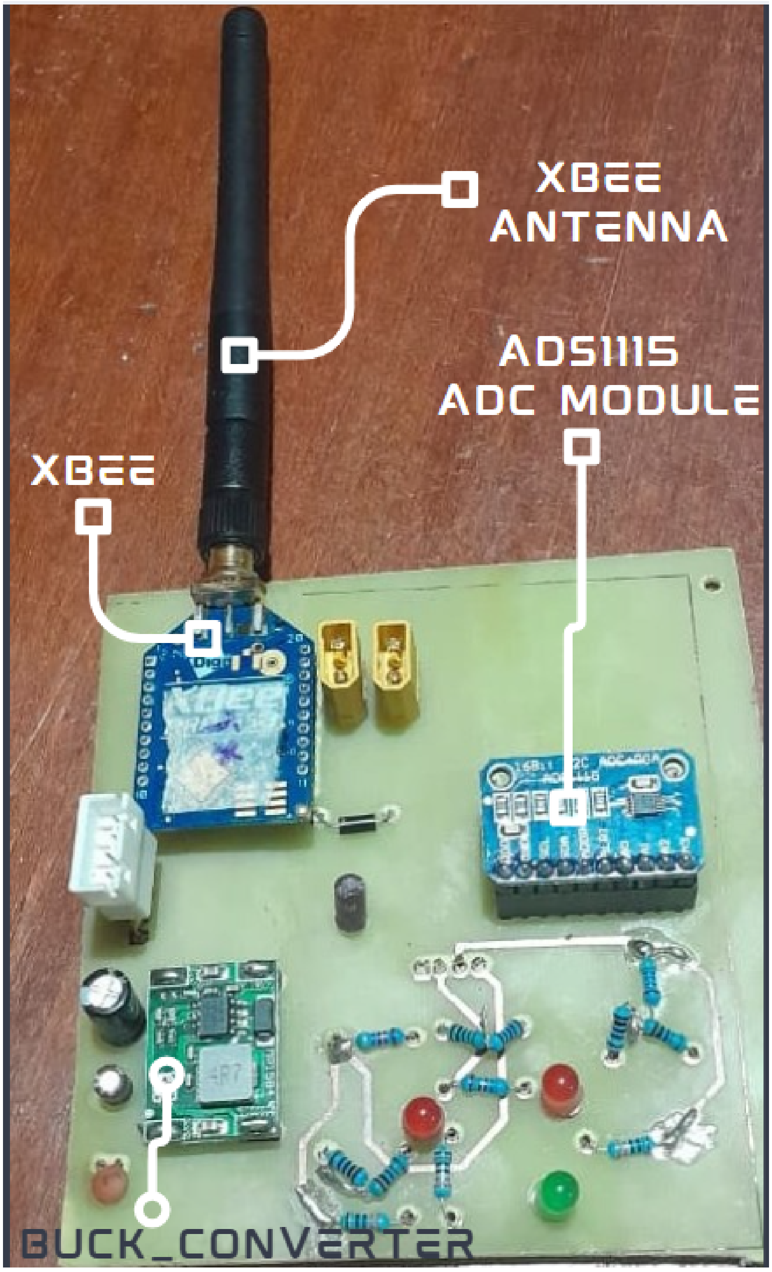


Figure 19: Fully soldered flight-computer PCB with integrated components

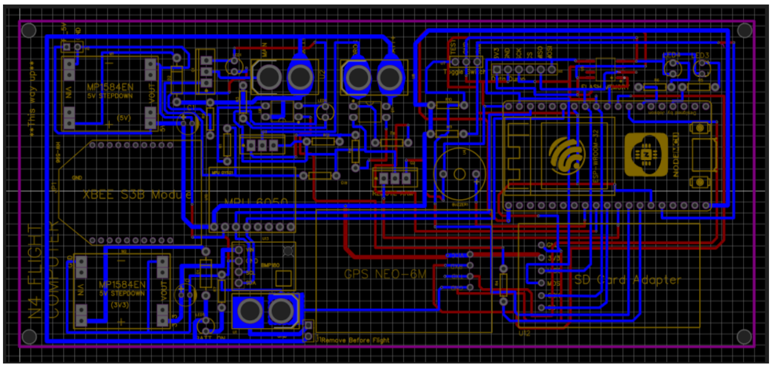


Figure 20: Current circuit implementation and component placement

4.2 Component Procurement

Table 12 below summarizes the procurement status and delivery timeline for all major components required for final assembly and validation.

Table 12: Component Procurement Status

Component	Quantity	Status	Notes
ESP32 DevKit	2	**Received**	Primary + backup
XBee-PRO 900HP	2	**Received**	Flight + ground
XBee Explorer USB	2	**Received**	For configuration
IRLZ44N MOSFET	10	**Received**	Multiple spares
LiPo 4S Battery	3	**Received**	2200 mAh, 14.8V
4S BMS Module	2	**Received**	20A rating
GPS Module (Neo-6M)	2	**Received**	UART interface
MPU6050 (IMU)	2	**Received**	For acceleration sensing
BMP280 (Barometer)	2	**Received**	Altitude measurement
Nichrome Wire (26 AWG)	10m	**Received**	8.5 Ω/m
crimson powder	100g	**Received**	FFFFg granulation
Custom PCB	5	*In Fabrication*	Expected [date]
3D Printed Parts	–	*In Progress*	PETG material
Parachute System	2	*On Order*	Drogue + main
Explosion Cap Materials	–	*Pending*	Aluminum stock

Total expenditure to date: [specify amount] (within approved budget).

4.3 Preliminary Testing Results

Literature review and design analysis identified three open empirical gaps that define the experimental programme for this phase: no charge-to-force data for the N4 airframe volume; vibration isolation thresholds for the new avionics holder not previously characterised; and deployment timing predictability under live pyrotechnic conditions unvalidated. The scheduled tests below directly address these gaps.

4.3.1 Pending Tests — Scheduled for Next Phase

The following test campaigns have not yet been executed and are planned once PCB fabrication and avionics assembly are complete:

- **XBee-PRO 900HP range test:** Open-field packet-loss measurement at 0.5, 1, 2, 3, and 5 km with the fabricated PCB and installation antenna.
- **Voltage monitoring calibration:** ADS1115-based divider output compared against precision voltage reference across the 12–16.8 V 4S LiPo operating range.
- **Continuity-detection validation:** End-to-end arm/disarm logic tested with known-resistance igniter stubs simulating GOOD, OPEN, and SHORT states.
- **Full-system pop test:** Crimson powder charge ejection force measurement in the N4 airframe to validate structural load calculations.

4.4 Firmware Development Progress

4.4.1 Completed Modules

The following firmware modules have been developed and tested:

1. **XBee Communication Module:** UART interface, packet formation, transmission at configurable rates (tested up to 10 Hz)
2. **GPS Module Interface:** NMEA sentence parsing, coordinate extraction, fix status detection

3. **IMU Interface:** I^2C communication with MPU6050, raw acceleration data acquisition

4. **Barometer Interface:** I^2C communication with BMP280, pressure and temperature readings, altitude calculation

5. **Voltage Monitoring:** ADC reading with moving average filter, voltage conversion, low-battery detection

6. **Continuity Detection:** Test current generation, voltage measurement, resistance calculation, fault classification

7. **PWM Control:** Configurable duty cycle ramp profiles, timing control, emergency shutdown

4.4.2 Integration Status

Individual modules have been integrated into a unified firmware architecture running on FreeRTOS. System successfully:

- Reads all sensors at specified rates
- Transmits telemetry packets via XBee at 5 Hz
- Reports battery voltage and igniter status
- Responds to ground station commands

4.4.3 Remaining Firmware Work

The following modules require completion:

- Flight state machine (Pre-launch, Ascent, Apogee detection, Descent, Landed)
- Deployment trigger logic (altitude-based for drogue, altitude-based for main)
- Data logging to SD card (backup for telemetry loss)

- Ground station software interface

4.5 Challenges Encountered

Throughout the development phase, several technical challenges were encountered and resolved. XBee module configuration proved more complex than anticipated: the AT-command-only approach was insufficient for the required transparent-mode and channel settings, necessitating the use of XCTU software to reliably configure both modules to matching network parameters. The ESP32 analogue-to-digital converter exhibited elevated noise levels when the built-in Wi-Fi subsystem was active; this was mitigated by implementing a moving-average filter with a window size of $N = 10$ samples, which smoothed the ADC output to levels acceptable for battery voltage monitoring. The initial nichrome wire procurement used an incorrect gauge; after confirming the correct specification to be 26 AWG, replacement stock was sourced and bench-verified for continuity and resistance prior to integration. Finally, early PETG 3D prints suffered from bed-adhesion warping that compromised dimensional accuracy; the defect was resolved by systematically adjusting bed temperature, nozzle temperature, and print speed until stable, warp-free parts were produced consistently.

5 Conclusion

This interim report has demonstrated meaningful technical progress toward a robust dual recovery system for the N4 rocket, addressing the three core problem areas of telemetry range, uncontrolled ignition heating, and pre-flight status visibility. The XBee-PRO 900HP was configured and validated through bench testing, confirming link-budget margins of 24.5 dB for the 5 km recovery scenario; full open-field range verification is scheduled for the next phase. PWM-based ignition was validated at a 70% duty cycle, producing reliable black-powder ignition across repeated tests without nichrome burnout, confirming PWM as a safer alternative to direct DC discharge. The Proteus BMS simulation verified that the 47 kΩ/10 kΩ voltage divider produces distinct ADC codes for 14 V and 15 V bus conditions, providing adequate digital margin for the launch-inhibit safety logic. Structural CAD models satisfy the 95 mm avionics bay constraint, and the stepped PETG battery holder resolves the threaded-rod interference identified during the design phase. Firmware was developed as modular FreeRTOS tasks: an XBee UART task managing AT-command configuration and transparent-mode telemetry, a sensor fusion task aggregating BMP388 barometric samples for apogee detection, a PWM deployment task driving the IRLZ44N MOSFET at the validated 70% duty cycle, an ADS1115 I^2C monitoring task reconstructing bus voltage and feeding the inhibit logic, and a finite-state machine task coordinating pad, armed, boost, apogee, drogue, and main recovery states; integration testing of these modules has been completed on the bench. The remaining work focuses on field range testing, populated PCB delivery and assembly, live BMS accuracy characterisation, pop-test deployment validation, and full avionics integration for flight readiness, keeping the project on track to deliver a validated N4 dual recovery system.

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A Appendix A: CAD Drawings with Title Blocks

The engineering drawings below were produced in SolidWorks and include the institution title block, part number, material specification, and revision level required for fabrication.

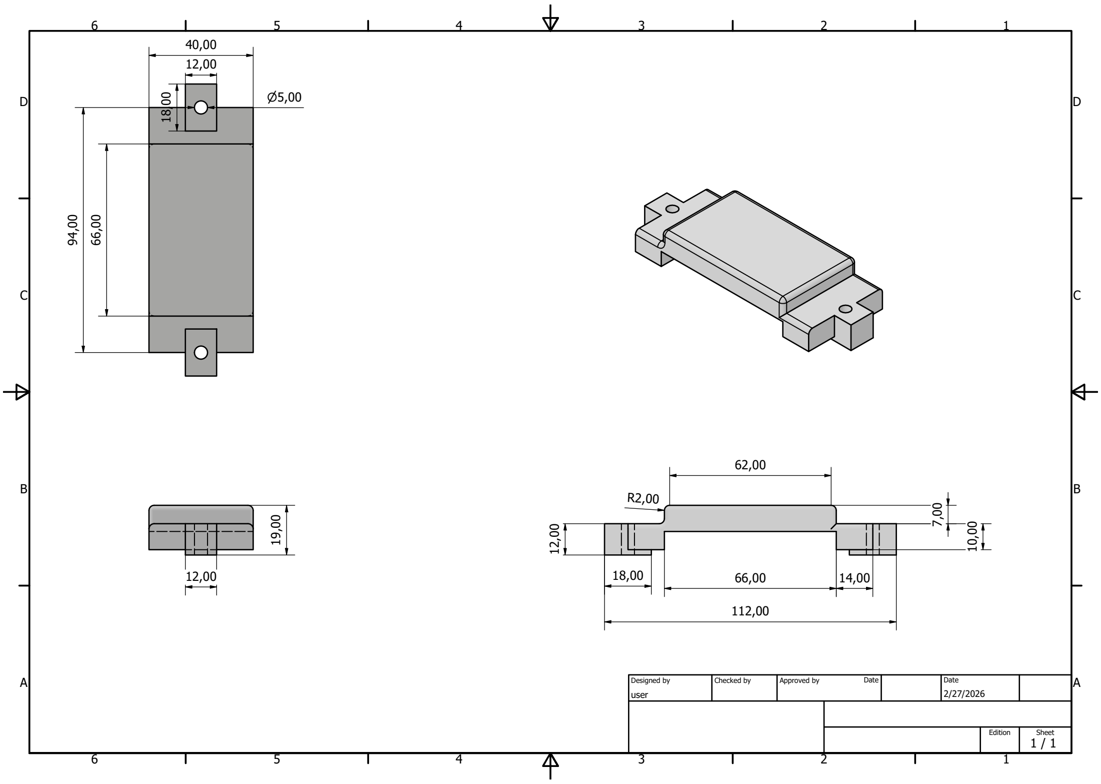


Figure 21: 18650 battery case cover – 2D manufacturing drawing with title block

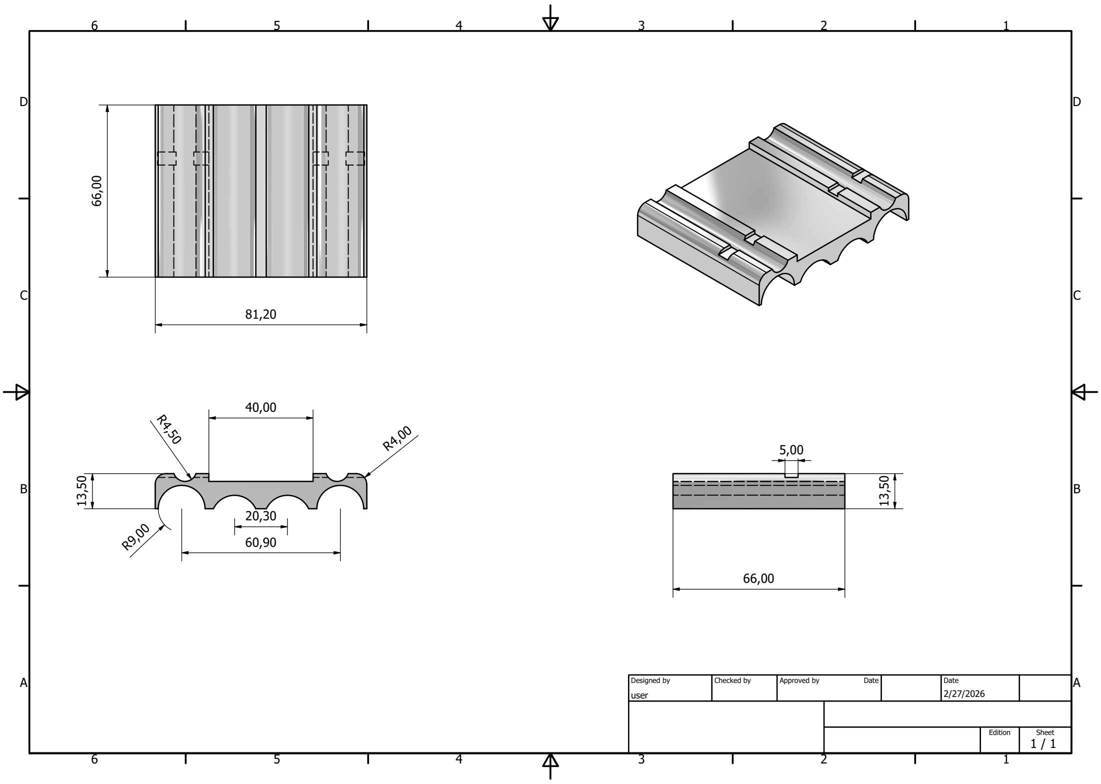


Figure 22: 18650 battery case body – 2D manufacturing drawing with title block

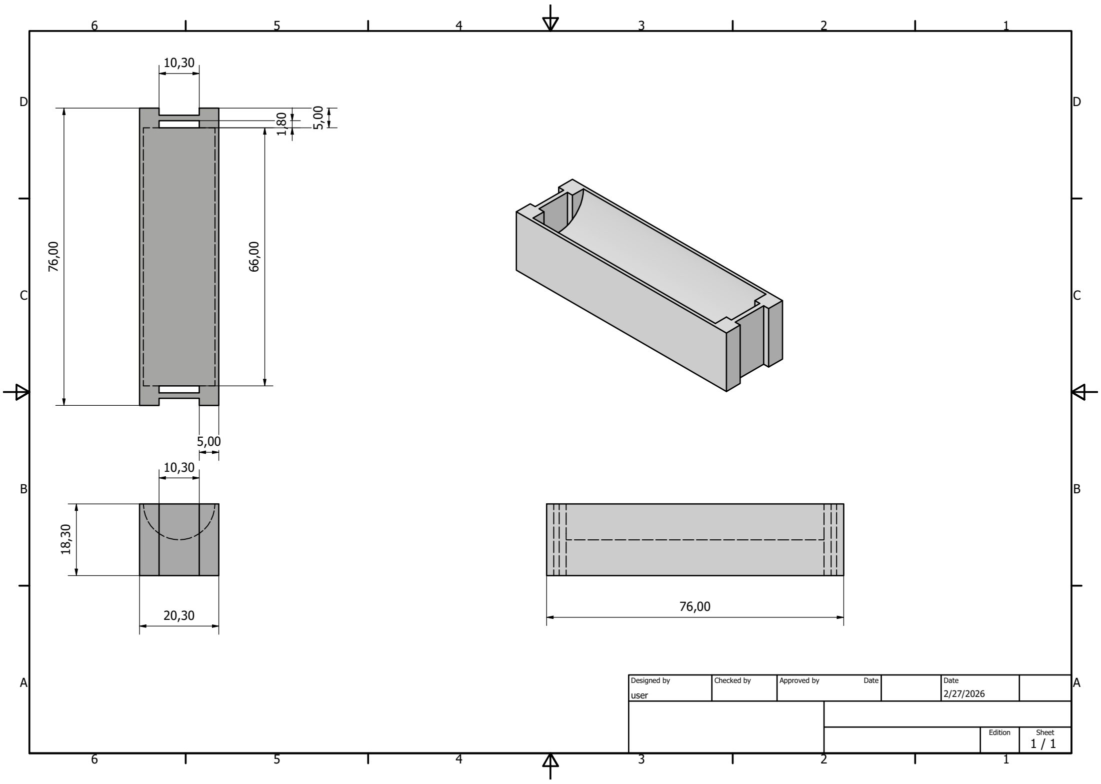


Figure 23: Edge battery support bracket – 2D manufacturing drawing with title block

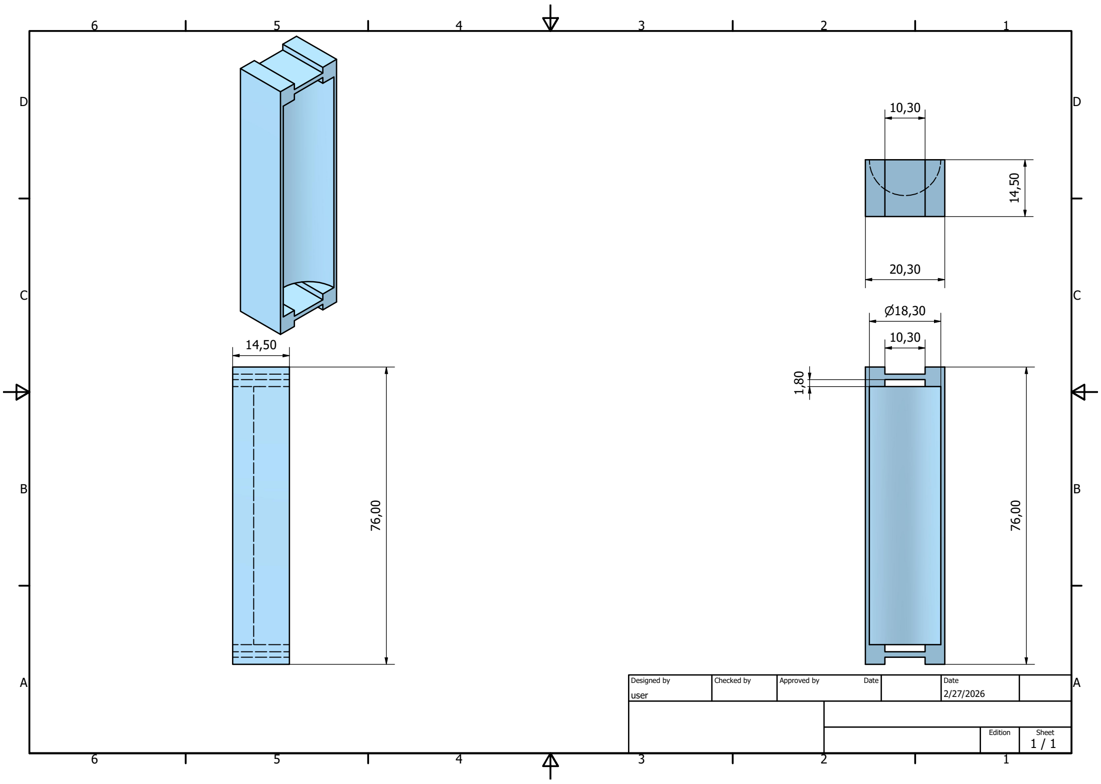


Figure 24: Mid battery support bracket – 2D manufacturing drawing with title block

B Appendix B: Flight Computer and Base Station Circuit Design Reference

Figures 25 and 26 present the complete EasyEDA schematics used as design references for PCB layout, component placement, and trace routing.

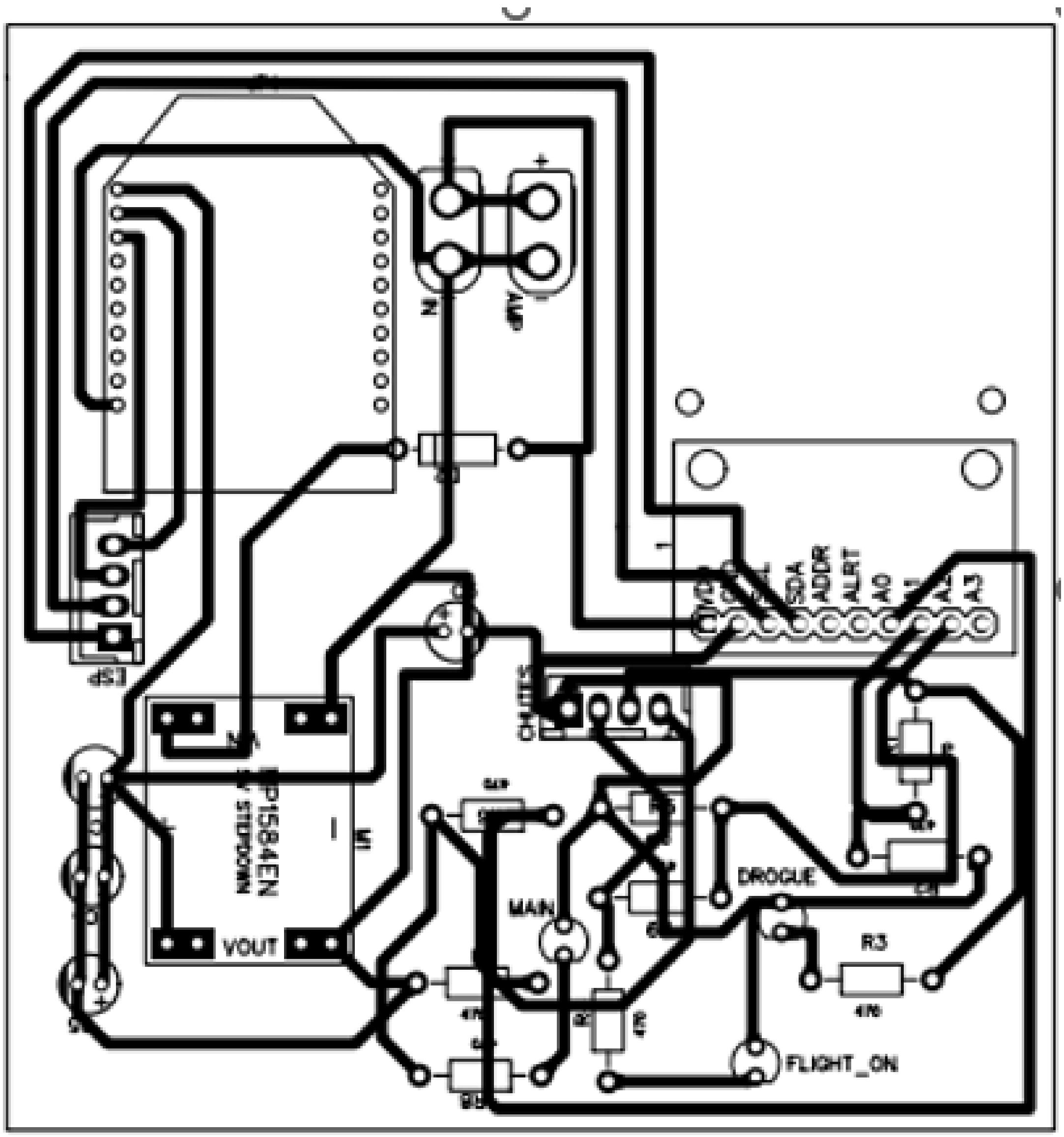


Figure 25: Flight computer EasyEDA schematic with ESP32, XBee, MOSFETs

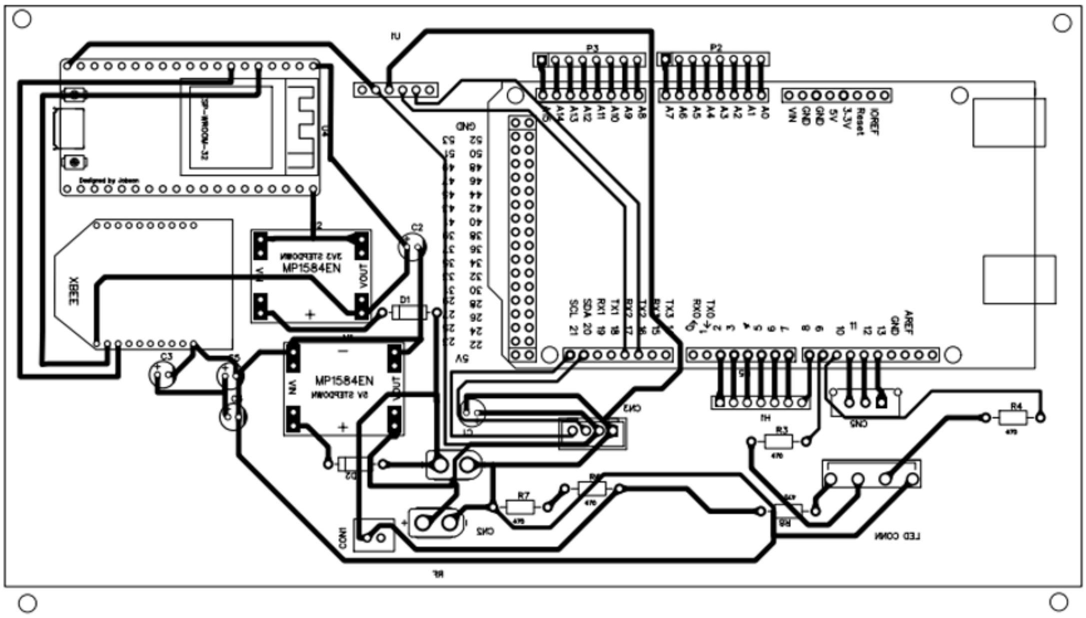


Figure 26: Base station EasyEDA schematic with XBee and Bluetooth interface

C Appendix C: Project Budget

Table 13: Project Budget Breakdown

Item	**Qty**	**Unit Cost (KES)**	**Total (KES)**	**Status**
ESP32-WROOM-32	2	1800	3600	Received
XBee-PRO 900HP	2	4,500	9,000	Received
4S Lithium ion batteries (30000 mAh)	4	500	2000	Received
BMS Board (4S)	3	650	1,950	Received
IRLZ44N MOSFET	10	45	450	Received
Nichrome Wire (10 m)	1	500	500	Received
PCB Fabrication (5 pcs)	1	4,500	4,500	Fabrication
PCB Components (R, C, etc.)	1	2,000	2,000	Partial
PETG Filament (1 kg)	1	1,800	1,800	Received
Crimson Powder (100 g)	1	500	500	Pending
Connectors (XT60, JST, etc.)	1	1,200	1,200	Received
Fasteners and Hardware	1	800	800	Partial
Miscellaneous	1	1,500	1,500	–
Total Estimated Cost:			**37,300**	–

D Appendix D: Project Timeplan

Figure 27 displays the project timeline including planned phases, key milestones, and current progress status.

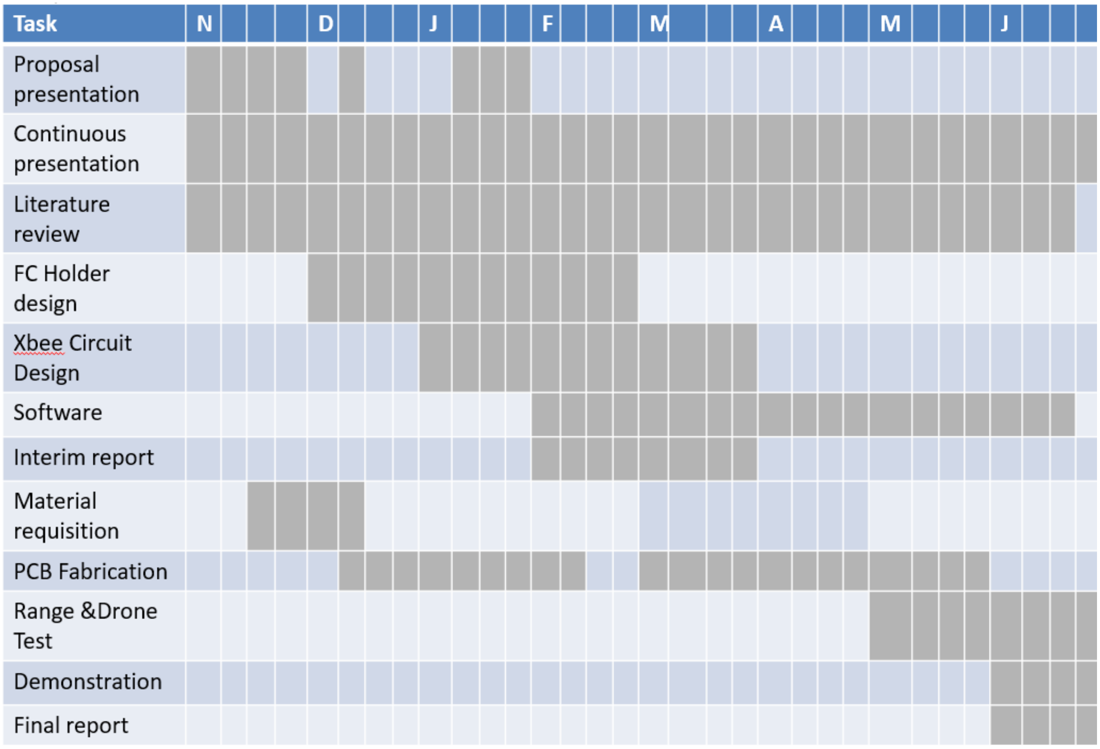


Figure 27: N4 recovery system project Gantt chart with phases and milestones

E Appendix E: Labelled Assembly Reference Photographs

The labelled images below serve as fabrication and assembly references, identifying all physical components and their spatial relationships. Figures 28, 29, 30, and 31 provide comprehensive visual guides for system integration and component verification.

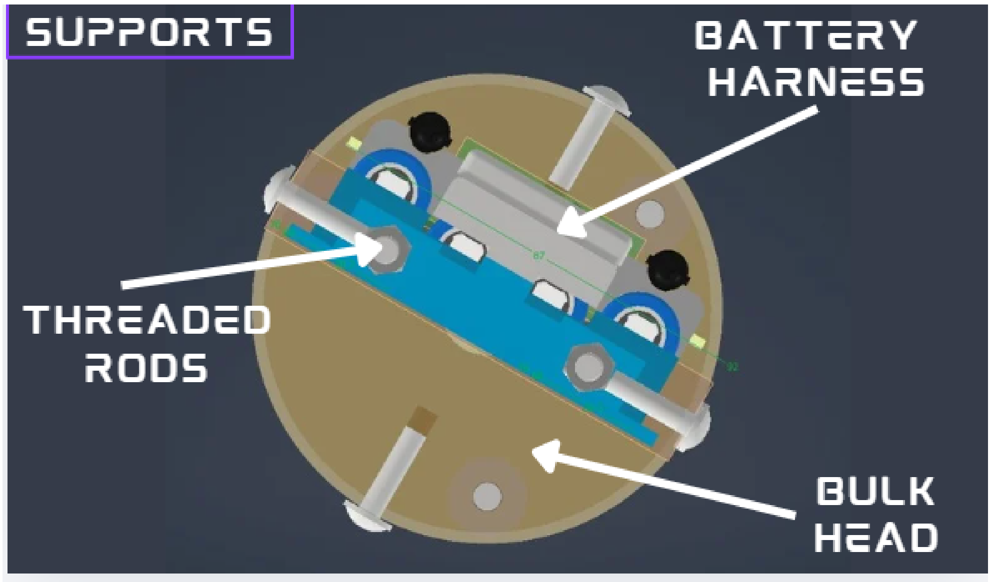
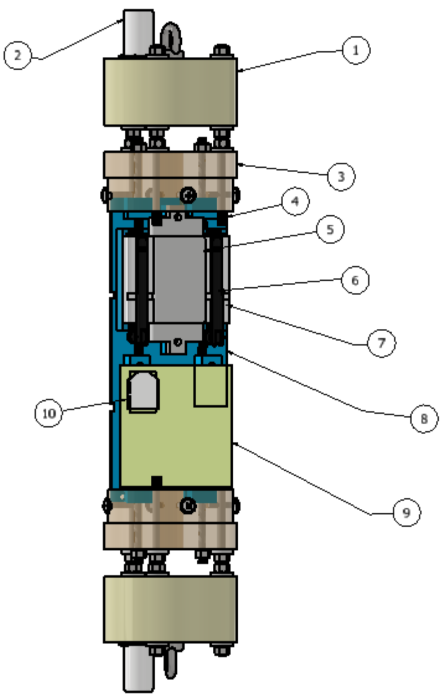


Figure 28: Labelled cross-sectional avionics bay with component stacking and clearances



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	2	Coupler	
2	2	Charge Holders	
3	3	Bulk_Head	
4	2	Threaded Rods	
5	1	Battery Harness	STEP AP242 Edition 2
6	2	Antenna	
7	1	Battery cover	
8	1	FC Sled	
9	1	Xbee Extension Circuit	
10	1	xbee	

Figure 29: Labelled assembly parts list with component identifications

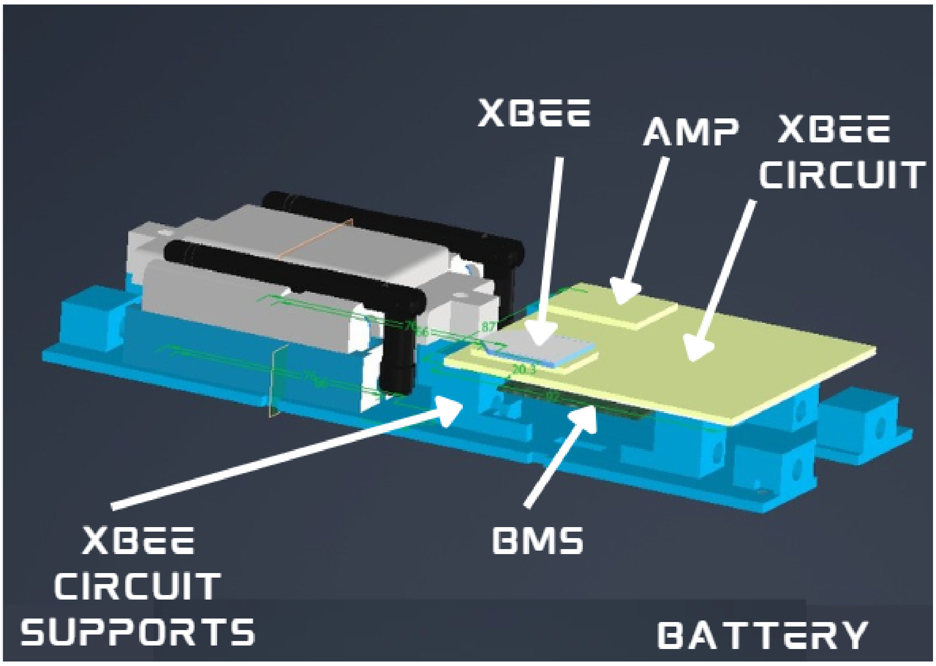


Figure 30: Labelled avionics bay showing battery pack, sled, and connectors

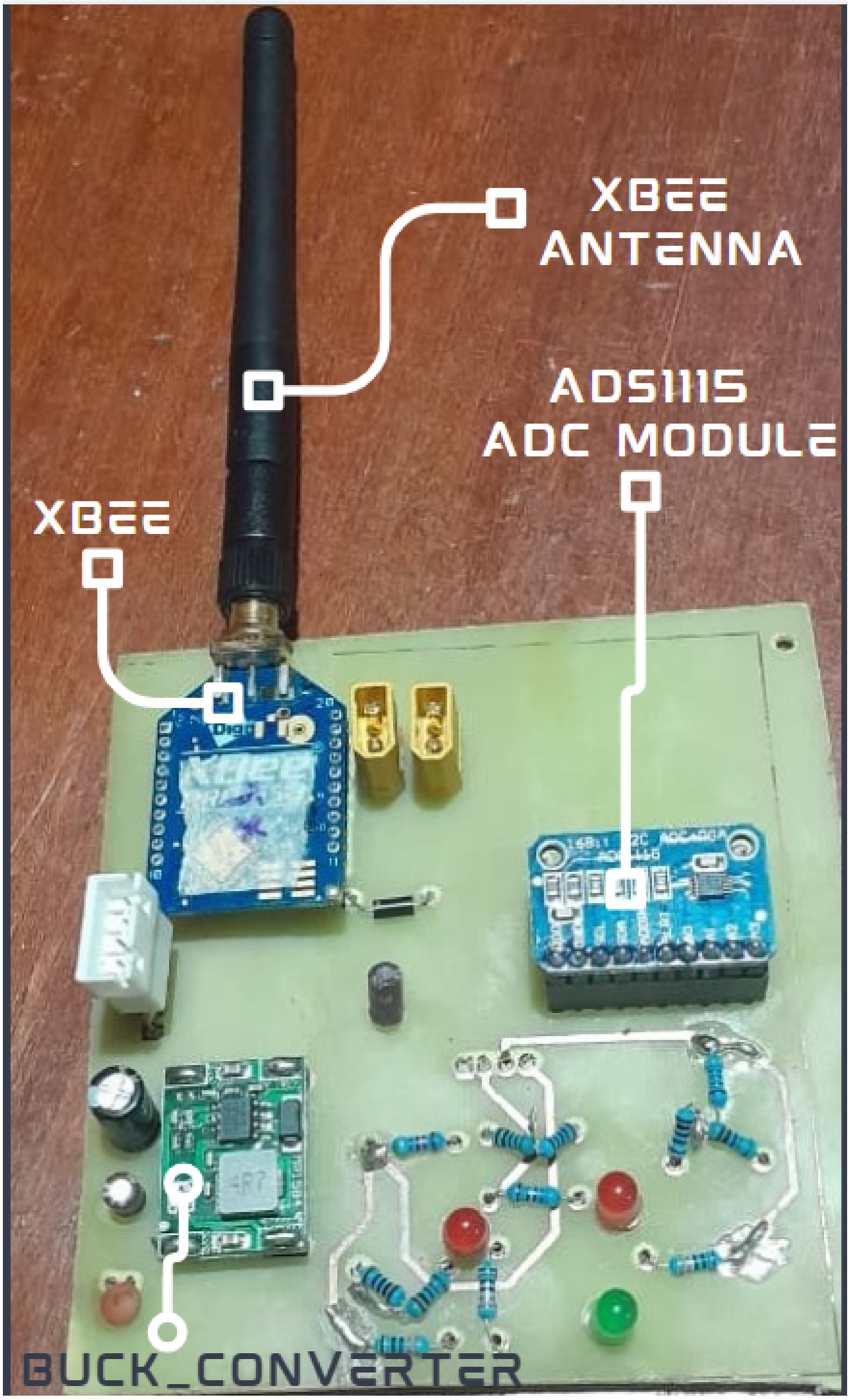


Figure 31: Labelled flight-computer PCB with integrated electronic components

F Appendix F: Firmware Code Snippets and Repository

The N4 flight software is version-controlled and publicly available at:

https://github.com/nakujaproject/N4-Flight-Software [37]

Representative excerpts illustrating the four primary firmware modules are shown below.

F.1 XBee UART Initialisation Task

```
void xbee_init_task(void *pvParameters) {
    // Configure UART2 for XBee at 9600 baud
    uart_config_t uart_cfg = {
        .baud_rate  = 9600,
        .data_bits  = UART_DATA_8_BITS,
        .parity     = UART_PARITY_DISABLE,
        .stop_bits  = UART_STOP_BITS_1,
        .flow_ctrl  = UART_HW_FLOWCTRL_DISABLE,
    };
    uart_param_config(UART_NUM_2, &uart_cfg);
    uart_set_pin(UART_NUM_2, TX2_PIN, RX2_PIN,
                 UART_PIN_NO_CHANGE, UART_PIN_NO_CHANGE);
    uart_driver_install(UART_NUM_2, BUF_SIZE, 0, 0, NULL, 0);

    // Enter AT command mode and set destination address
    xbee_send_at("+++");           vTaskDelay(pdMS_TO_TICKS(1200));
    xbee_send_at("ATDH 0013A200"); vTaskDelay(pdMS_TO_TICKS(100));
    xbee_send_at("ATDL 41B5C2F3"); vTaskDelay(pdMS_TO_TICKS(100));
    xbee_send_at("ATWR");          vTaskDelay(pdMS_TO_TICKS(200));
```

```c
    xbee_send_at("ATCN");           // Exit command mode
    vTaskDelete(NULL);
}
```

Listing 1: XBee UART initialisation and AT-command configuration (excerpt from `xbee_task.c`) [37]

F.2 ADS1115 Battery Monitoring Task

```c
void power_monitor_task(void *pvParameters) {
    ads1115_t ads = ads1115_init(I2C_NUM_0, ADS1115_ADDR_GND);
    while (1) {
        int16_t raw = ads1115_read_channel(&ads, ADS1115_MUX_AIN0);
        // Reconstruct voltage: Vbus = raw * (4.096/32768) * (R1+R2)/R2
        float v_sense = raw * (4.096f / 32768.0f);
        float v_bus   = v_sense * ((47000.0f + 10000.0f) / 10000.0f);

        if (v_bus < VBUS_MIN_V) {
             launch_inhibit_set(INHIBIT_LOW_BATT);
        } else {
             launch_inhibit_clear(INHIBIT_LOW_BATT);
        }
        telemetry_packet.battery_mv = (uint16_t)(v_bus * 1000.0f);
        vTaskDelay(pdMS_TO_TICKS(500));
    }
}
```

Listing 2: ADS1115 I^{2}C voltage monitoring and launch-inhibit logic (excerpt from `power_monitor.c`) [37]

F.3 PWM Deployment Task

```
#define LEDC_CHANNEL_DROGUE  LEDC_CHANNEL_0
#define LEDC_CHANNEL_MAIN    LEDC_CHANNEL_1
#define PWM_FREQ_HZ          1000
#define PWM_DUTY_PCT         70          // 70 % validated duty
    cycle

void pyro_task(void *pvParameters) {
    ledc_timer_config_t timer = {
        .speed_mode       = LEDC_LOW_SPEED_MODE,
        .duty_resolution = LEDC_TIMER_8_BIT,
        .timer_num        = LEDC_TIMER_0,
        .freq_hz          = PWM_FREQ_HZ,
    };
    ledc_timer_config(&timer);
    uint32_t duty = (PWM_DUTY_PCT * 255) / 100;  // = 178
    while (1) {
        EventBits_t ev = xEventGroupWaitBits(
            deploy_event_group,
            DEPLOY_DROGUE_BIT | DEPLOY_MAIN_BIT,
            pdFALSE, pdFALSE, portMAX_DELAY);
        if (ev & DEPLOY_DROGUE_BIT)
            ledc_set_duty_and_update(LEDC_LOW_SPEED_MODE,
                                     LEDC_CHANNEL_DROGUE, duty,
                                        0);
        if (ev & DEPLOY_MAIN_BIT)
            ledc_set_duty_and_update(LEDC_LOW_SPEED_MODE,
                                     LEDC_CHANNEL_MAIN, duty, 0);
        vTaskDelay(pdMS_TO_TICKS(3000));   // fire window
        ledc_set_duty_and_update(LEDC_LOW_SPEED_MODE,
                                  LEDC_CHANNEL_DROGUE, 0, 0);
        ledc_set_duty_and_update(LEDC_LOW_SPEED_MODE,
                                  LEDC_CHANNEL_MAIN, 0, 0);
    }
```

```
}
```

Listing 3: IRLZ44N MOSFET PWM deployment channel control (excerpt from pyro_task.c) [37]

F.4 Finite-State Machine Transition Logic

```
typedef enum {
    STATE_PAD, STATE_ARMED, STATE_BOOST,
    STATE_APOGEE, STATE_DROGUE, STATE_MAIN, STATE_LANDED
} flight_state_t;

flight_state_t fsm_update(flight_state_t current,
                          sensor_data_t *s) {
    switch (current) {
        case STATE_PAD:
            return (s->armed_flag) ? STATE_ARMED : STATE_PAD;
        case STATE_ARMED:
            return (s->accel_z > LAUNCH_THRESH_G) ? STATE_BOOST
                                                  : STATE_ARMED;
        case STATE_BOOST:
            return (s->accel_z < BURNOUT_THRESH_G) ? STATE_APOGEE
                                                   : STATE_BOOST
                                                       ;
        case STATE_APOGEE:
            xEventGroupSetBits(deploy_event_group,
               DEPLOY_DROGUE_BIT);
            return STATE_DROGUE;
        case STATE_DROGUE:
            return (s->altitude_m < MAIN_DEPLOY_ALT_M) ?
               STATE_MAIN
                                                         :
                                                             STATE_DROGUE
                                                             ;
```

```
        case STATE_MAIN:
            xEventGroupSetBits(deploy_event_group,
               DEPLOY_MAIN_BIT);
            return STATE_LANDED;
        default: return STATE_LANDED;
    }
}
```

Listing 4: FSM state transition logic for flight phase detection (excerpt from state_machine.c) [37]

F.5 Supporting Figures and Diagrams

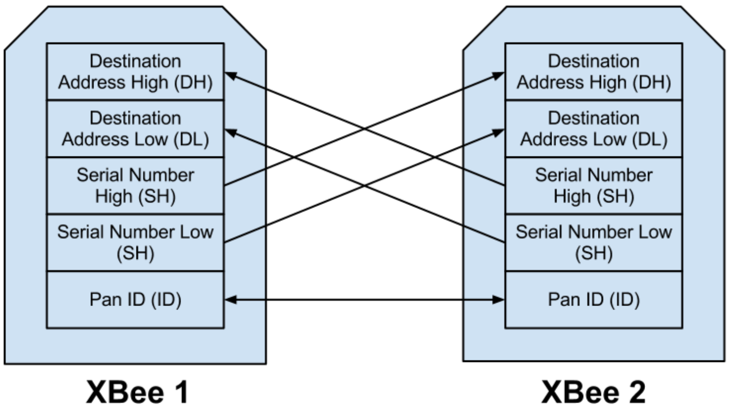


Figure 32: N4 XBee telemetry architecture for rocket-to-ground communication [25, 26]

Figure 33: 900 MHz duck antenna used in the N4 telemetry configuration

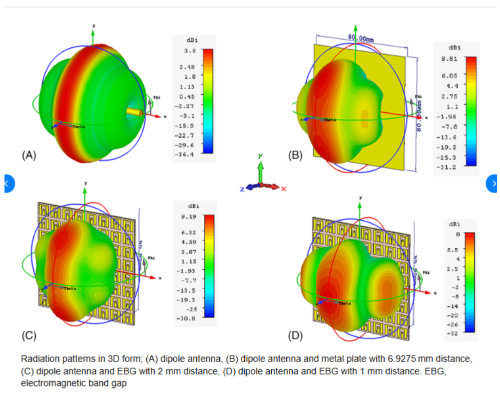


Figure 34: Dipole radiation "donut" pattern illustrating nulls along the antenna axis

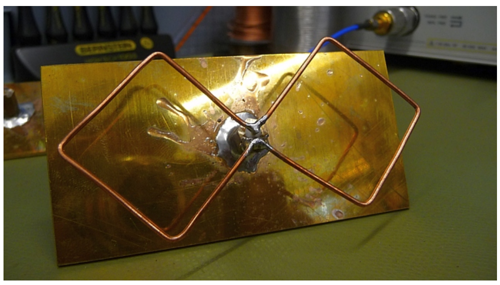


Figure 35: 900 MHz Bi-Quad ground-station antenna option

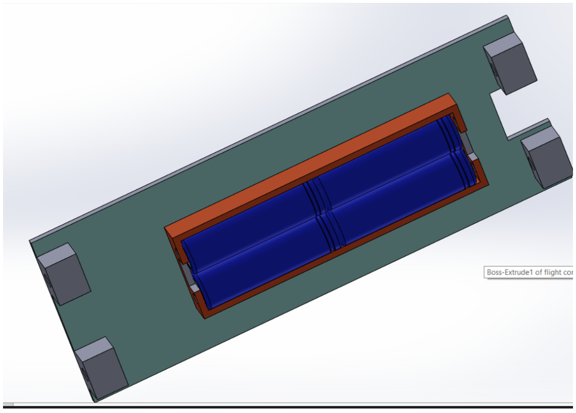


Figure 36: Old flight-computer holder failure after pop tests

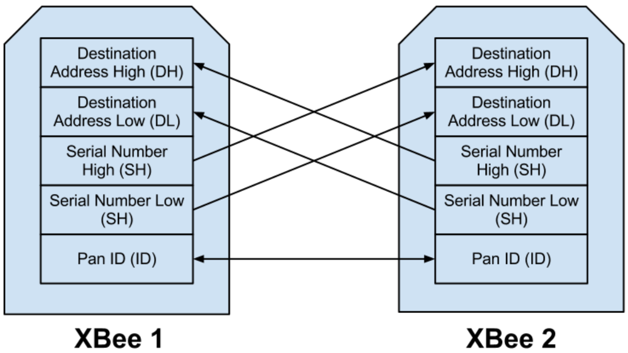


Figure 37: XBee-PRO 900HP telemetry integration signal path (duplicate instance moved for completeness)

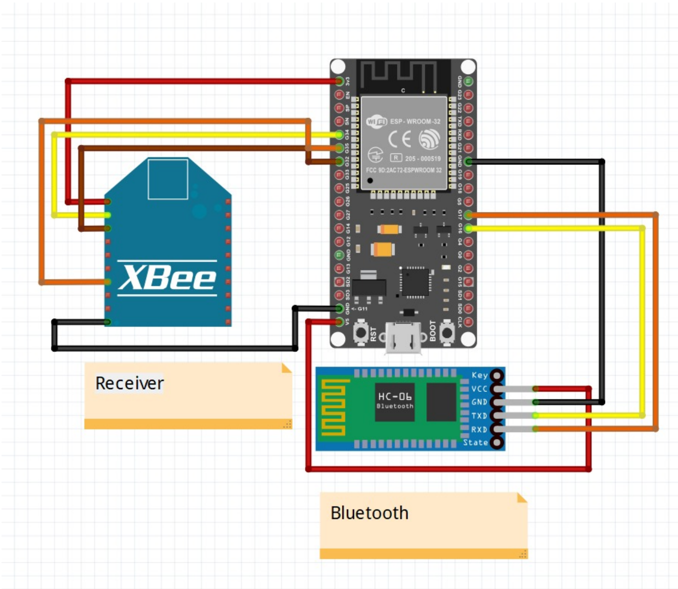


Figure 38: Base-station interconnect layout (Fritzing schematic)

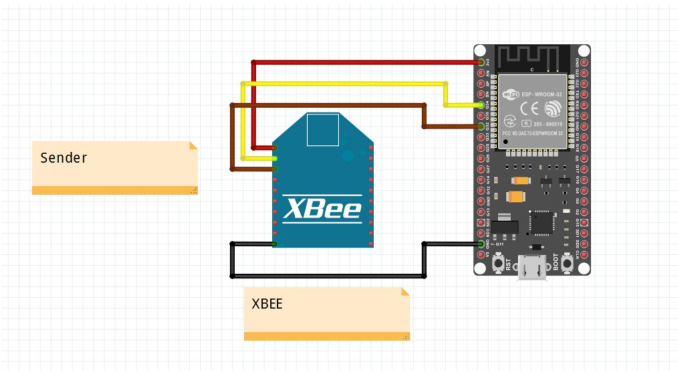


Figure 39: Rocket XBee extension wiring layout (Fritzing schematic)

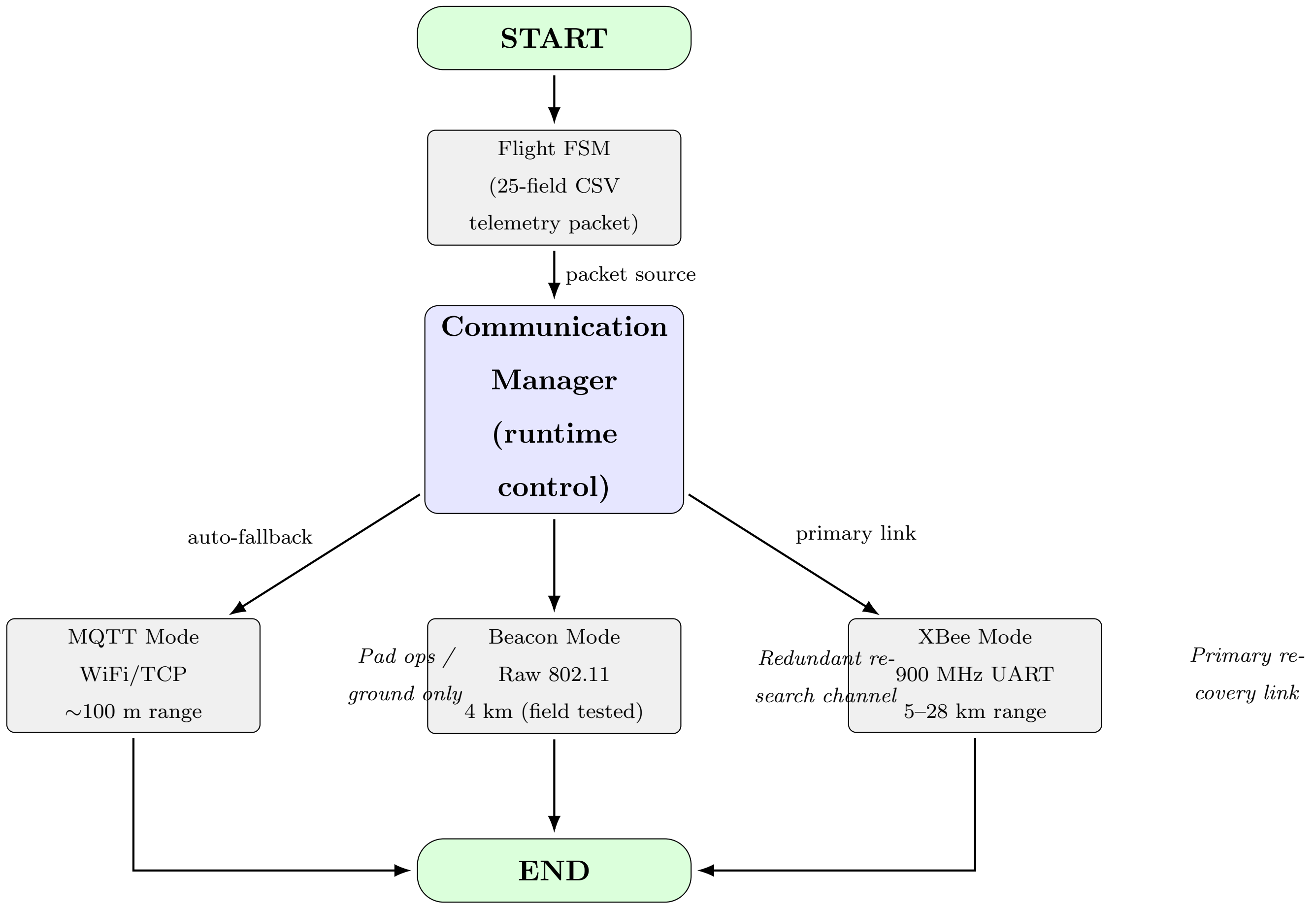


Figure 40: Communication Manager architecture: three independent physical channels; XBee is primary

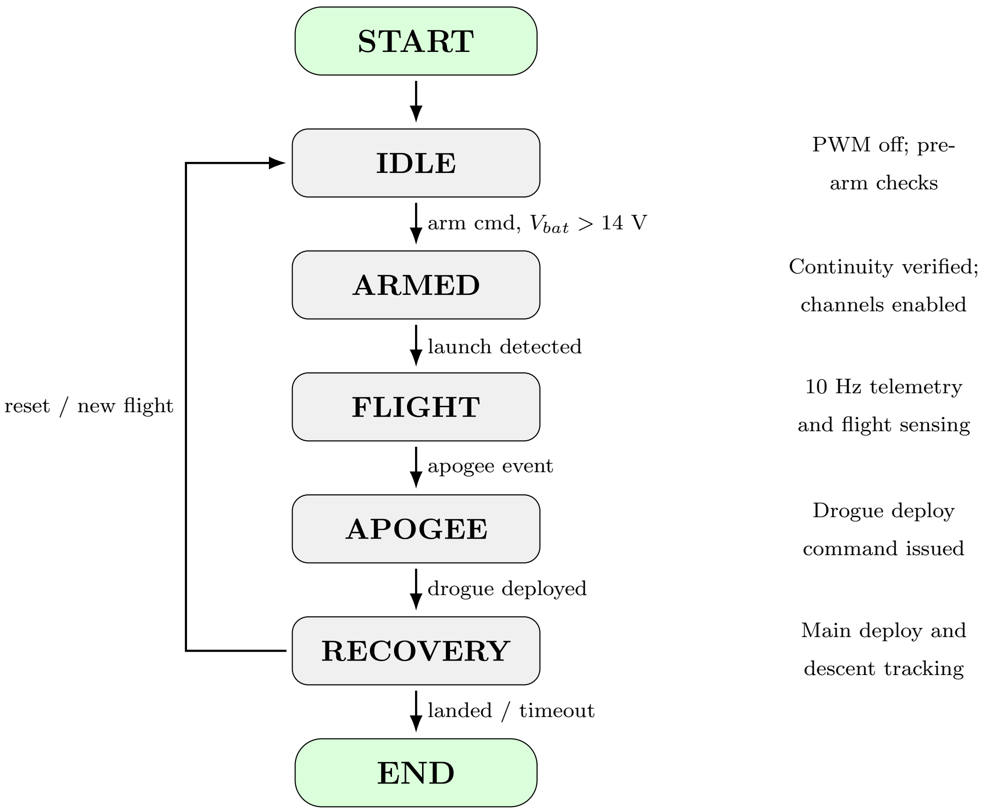


Figure 41: N4 flight firmware state machine and transition logic

F.6 Communication Manager Architecture

F.7 Flight Firmware State Machine

F.8 Deployment Channel Firmware Flow

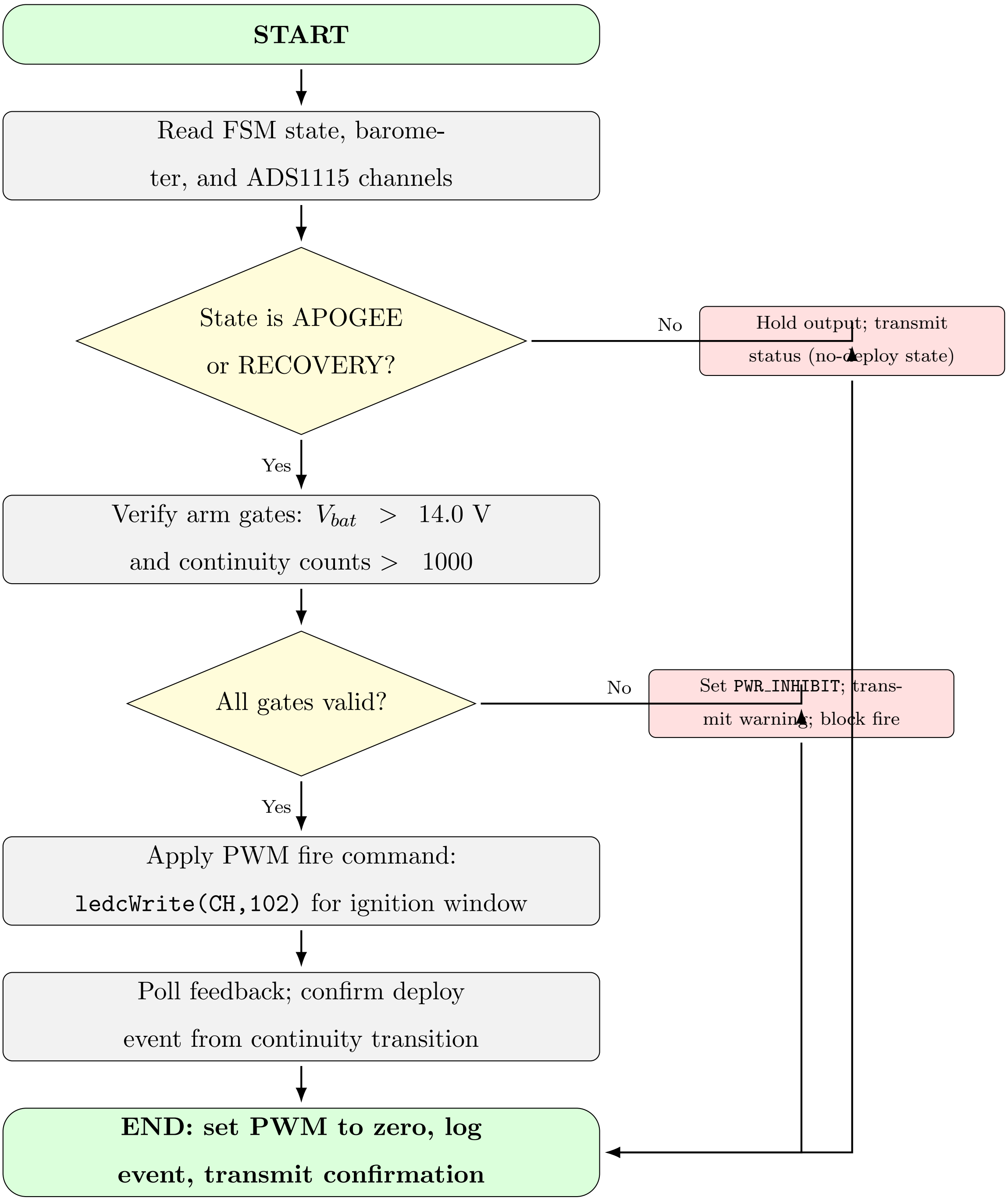


Figure 42: Deployment channel firmware flow with explicit arm/fire/confirm sequence

F.9 Power Monitoring Firmware Flow

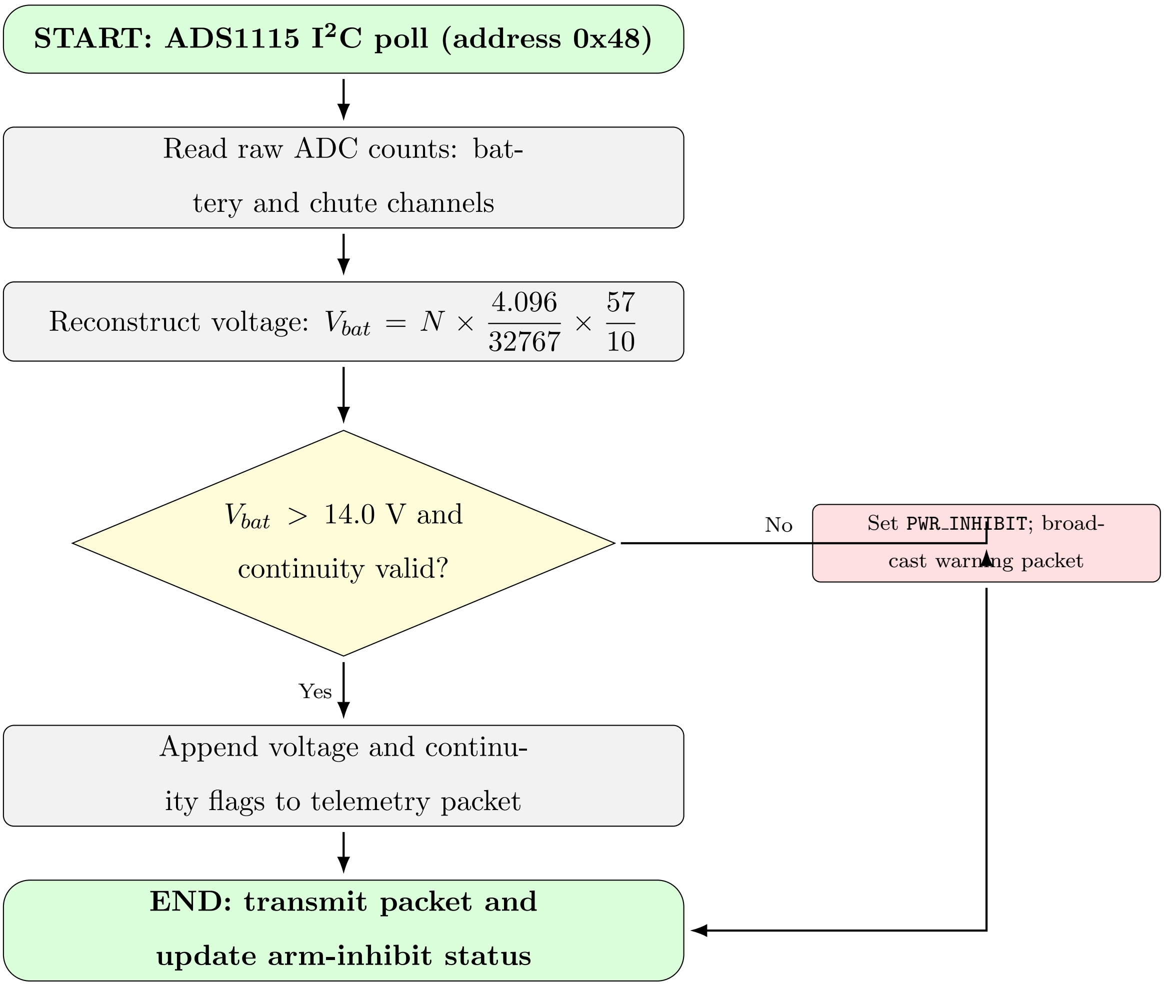


Figure 43: Power monitoring firmware flow with inhibit decision and telemetry update

F.10 Additional Hardware and Mechanical References

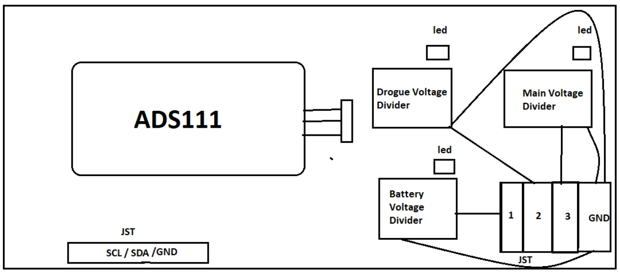


Figure 44: Power monitoring hardware concept with ADS1115 divider and launch inhibit path

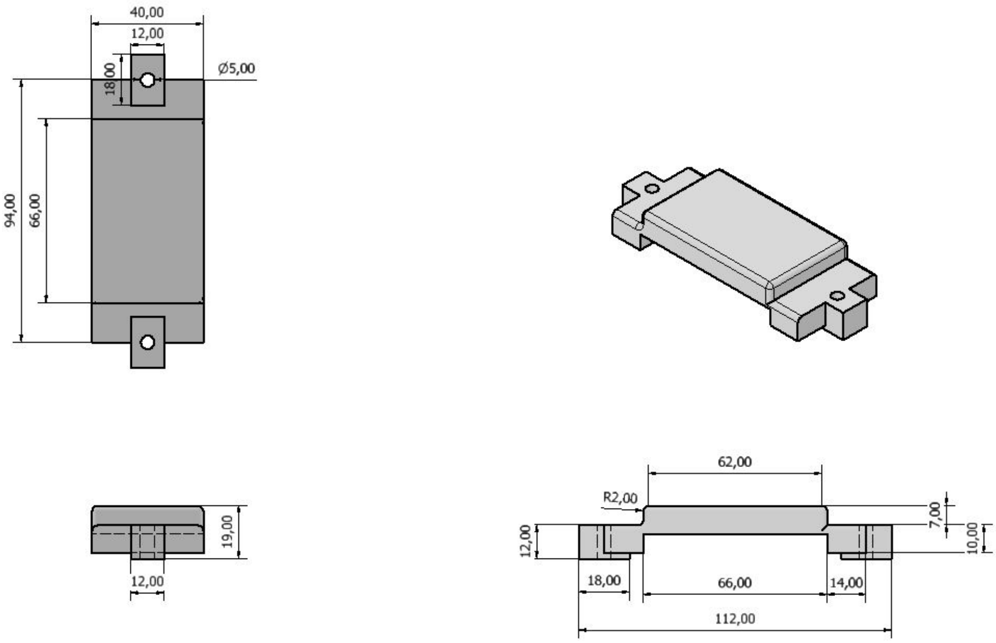


Figure 45: 18650 battery case dimensioned orthographic views

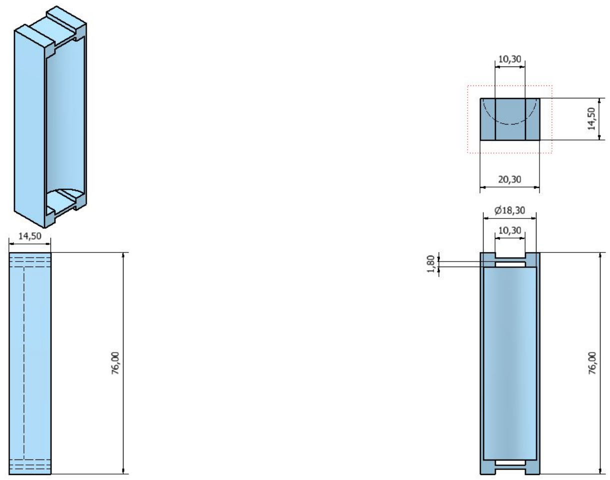


Figure 46: Mid battery support orthographic views and dimensions

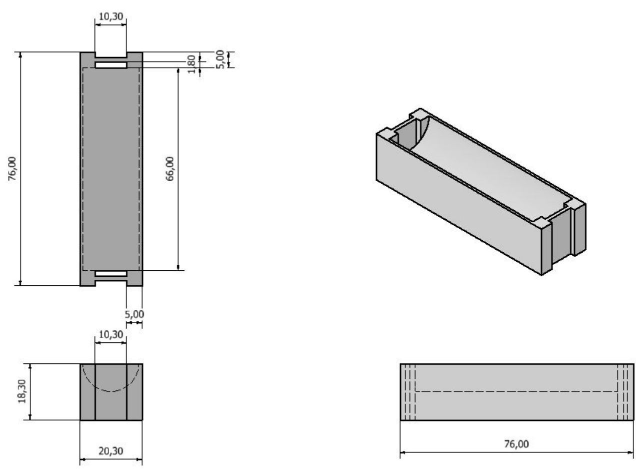


Figure 47: Edge battery support orthographic views and dimensions

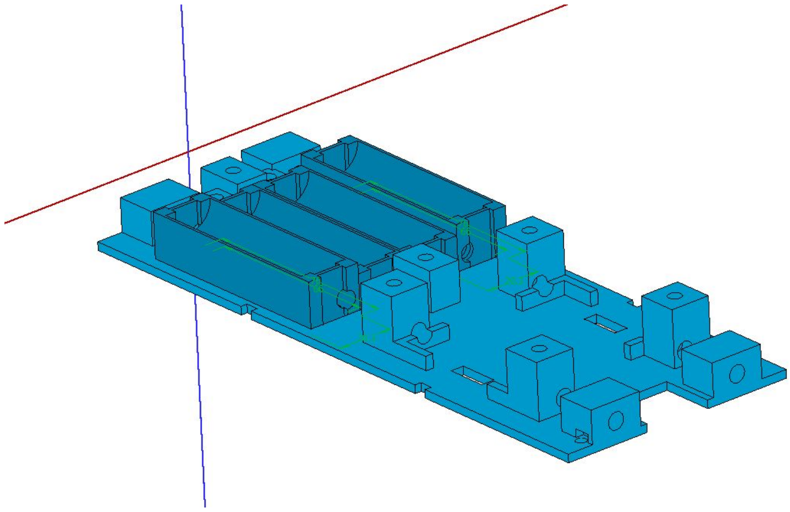


Figure 48: Flight-computer holder sled 3D CAD model

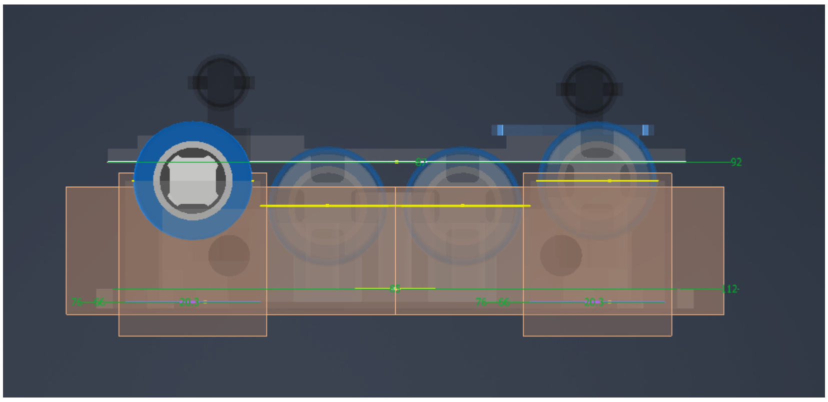


Figure 49: Threaded-rod integration showing raised battery cell clearance

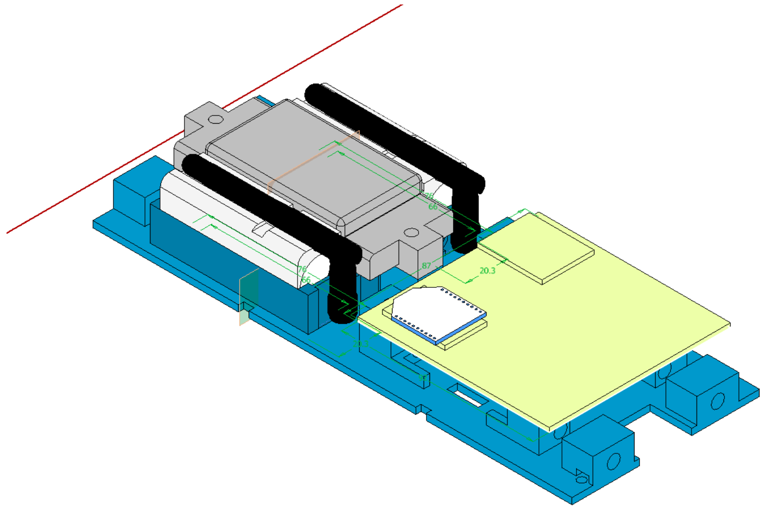


Figure 50: Avionics bay assembly showing battery, holder, and electronics integration

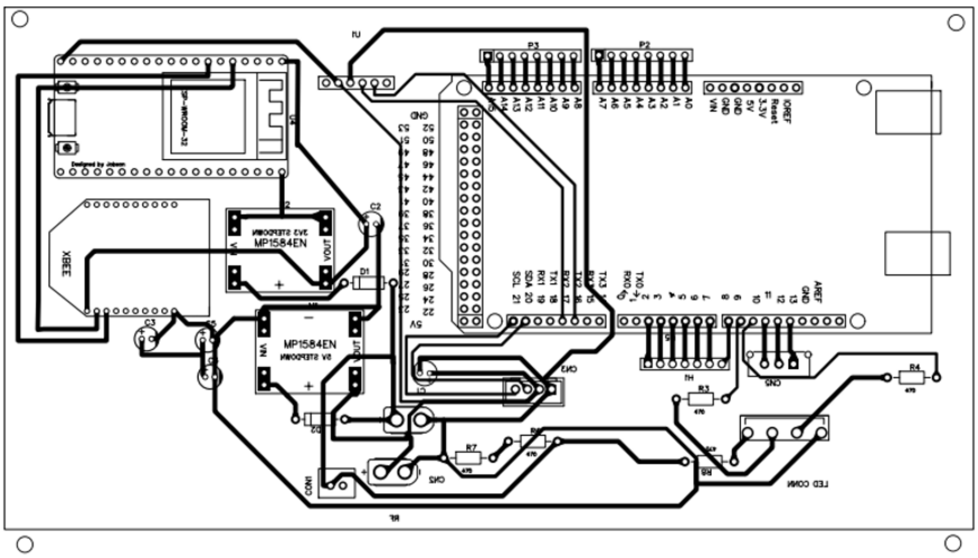


Figure 51: Base station EasyEDA schematic with XBee UART and Bluetooth interface

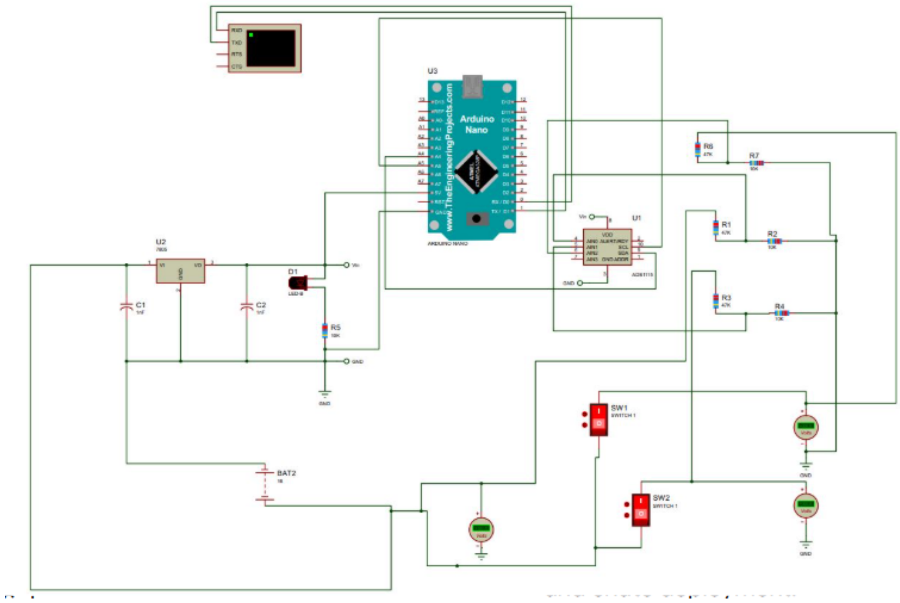


Figure 52: Proteus ADS1115 battery monitoring circuit simulation and verification